FemboyEngine2D API Reference Manual

C++26 data-oriented engine - complete public surface, contracts and a build-from-zero walkthrough

Version 0.4.0-alpha.1 (Alpha) - 2026-11-12 - English

Contents

- 1. Scope, conformance and engine rules
- 2. Building, targets and compiler flags
- 3. Core value types, FORCEINLINE and the cast family
- 4. FrameArena: the only frame allocator
- 5. JobSystem and per-thread diagnostics
- 6. ECS, scheduling and snapshots
- 7. RHI, render items and frame statistics
- 8. Materials: cooked graphs, the pipeline cache and the shipped path
- 9. Physics: Box2D v3, joints, cloth and SdfCollisionSystem
- 10. NetRollbackSession: deterministic rollback
- 11. Audio: mixer, DSP graph, limiter
- 12. PinkUI and WidgetCanvasSystem
- 13. Accessibility kernel (BlindMode, DeafMode, FlashGuard)
- 14. Asset pipeline, VFS and localization
- 15. Platform services: Steam, Epic Games Store, GOG
- 16. Editor: dashboard, fonts, LogBoy, Scene Diagnostics
- 17. Step-by-step: a game from an empty folder
- 18. Testing, style gate and release checklist
- 19. Open work
- 20. Appendix A - generated symbol index
- 21. Appendix B - macros and platform defines

1. Scope, conformance and engine rules

This manual specifies the public surface of FemboyEngine2D: every header under Source/Engine and the editor infrastructure that a host links against. It is written against the release-candidate tree, so a symbol described here exists in the headers that ship with that tree, and the appendix index is generated from those headers on every run.

The engine targets C++26 with GCC 16.2.0 or newer and Clang 23.1.4 or newer; the Windows release candidate is built with clang-cl against the MSVC standard library. Four rules are load bearing and every public API is designed around them:

- Simulation state is plain, trivially copyable data. No virtual calls, no owning pointers and no std::string belong in a component, snapshot or network payload.
- Hot paths do not allocate. Frame scratch memory comes from FrameArena, logs go through fixed char buffers, and callbacks are function pointers plus void*, never std::function.
- Determinism is a property of the data, not of the scheduler. Every subsystem exposes a checksummed image, and NetRollbackSession can only replay what it can reproduce byte for byte.
- No RTTI in engine code: dynamic_cast and typeid are rejected by the style gate. Cast<> walks stable type ids instead (see section 3).

The style gate (Source/Tools/Dev/CheckStyle.py) enforces the naming and structure rules: PascalCase file stems, Allman braces, braced control bodies, In-prefixed parameters, Scoped-prefixed locals, no trailing line comments and no Cyrillic in engine sources.

2. Building, targets and compiler flags

One CMake tree produces three kinds of artefact: the engine static library (FemboyEngine2D::Engine), the editor library and application (FemboyEngine2D::Editor), and the offline tools. A shipped game links the engine only; Dear ImGui, ImGuizmo, ImPlot and the TOML writer never enter a shipping target.

CMake option	Default	Effect
FBY_BUILD_EDITOR	ON	Builds fby_editor, the editor app and the panel plugins.
FBY_BUILD_TESTS	OFF	Builds the ctest suite under Source/Tests.
FBY_BUILD_TOOLS	ON	FempakCooker, LocTool, SceneTool, ShaderCompiler, AotCompiler.
FBY_BUILD_PLATFORM_PROVIDERS	ON	Steam and Epic Games Store provider plugins.
FBY_RHI_BACKEND	auto	auto, vulkan, metal or null. auto selects per platform.

```
cmake -S . -B build -G Ninja -DCMAKE_BUILD_TYPE=RelWithDebInfo \
      -DFBY_BUILD_TESTS=ON -DFBY_BUILD_EDITOR=ON
```

```
cmake --build build --target fby_editor
ctest --test-dir build --output-on-failure
python Source/Tools/Dev/CheckStyle.py
```

The compiler wrapper (cmake/FbyCompiler.cmake) owns the C++26 flag, the warning set and the sanitizer switches. Do not add flags per target: a target that needs a different dialect is a target that should not exist.

3. Core value types, FORCEINLINE and the cast family

Definitions.hpp is the contract header of the engine. It supplies the export macros, the assertion macros (FBY_ASSERT, FBY_VERIFY, FBY_UNREACHABLE), the stable TypeId machinery, the cast family and the component registration macros. Every other header includes it, directly or transitively.

```
#if defined(_MSC_VER)
#define FORCEINLINE __forceinline
#elif defined(__GNUC__) || defined(__clang__)
#define FORCEINLINE inline __attribute__((always_inline))
#else
#define FORCEINLINE inline
#endif
```

FORCEINLINE is the only inlining spelling in public headers. It is used for accessors, hot arithmetic and template casts; it never replaces a measured optimisation. The three families are:

Symbol	Contract
FemCast<To>(value)	Unchecked static_cast with reference collapsing. Use when the concrete type is proven by construction (factory output, storage tag). No runtime cost.
Cast<To>(object)	Checked cast. Upcast is a compile-time static_cast; downcast compares the object's GetTypeId() against StableTypeId<To>() and its registered parent chain; unrelated types fail to compile. Returns nullptr on a mismatch, never throws, never uses RTTI.
CastComponent<To>(registry, entity)	ECS spelling of Cast<>: the type system proves the component type at compile time and the registry validates the entity. Returns nullptr when the entity lacks the component.

```
// Downcast through the stable type table, not through RTTI.
Fby::TypeObject* base = factory.Create("Sprite");
if (Fby::Sprite* sprite = Fby::Cast<Fby::Sprite>(base))
{
    sprite->Animate(dt);
}
```

```
// Registration binds Sprite to TypeObject and derives its stable id.
FBY_REGISTER_TYPE(Fby::Sprite, Fby::TypeObject);
```

StableTypeId<T>() is a consteval FNV-1a of the type's registered name (or the compiler signature when no name is pinned). A type that changes its registered name changes its id, which is why component names are part of the serialized schema and why Snapshot v5 stores a LayoutHash next to the name.

4. FrameArena: the only frame allocator

FrameArena is a bump pointer over caller-owned storage. It is reset at the start of a frame and never frees individual allocations. Systems that need transient arrays take it from SystemContext::frameArena; a system that calls operator new in its frame path is a defect, not a style preference.

```
Fby::FrameArena arena(storage);          // caller owns storage
arena.Reset();                           // once per frame
auto* items = arena.AllocateArray<MeshRenderItem>(count);
if (items == nullptr)
{
    // Arena exhausted: degrade, never allocate on the heap.
    return;
}
const std::size_t used = arena.Used();
const std::size_t capacity = arena.Capacity();
```

- Allocate() returns nullptr instead of throwing when the arena is exhausted; callers must handle the failure.
- AllocateArray<T>() static_asserts std::is_trivially_copyable_v<T>: the arena stores POD payloads only.
- Over-aligned types must pass their alignment explicitly; the default is alignof(std::max_align_t).

5. JobSystem and per-thread diagnostics

JobSystem owns a pool of std::jthread workers. Tasks are function pointers plus a context pointer plus a half-open range, so submitting a task never allocates. ParallelFor splits a range into chunks, lets every participant pull the next chunk from an atomic counter, and always runs at least one chunk on the caller, which keeps single-core machines correct and worker re-entry safe.

```
jobs.ParallelFor(entityCount, 4096, [&](std::uint32_t begin, std::uint32_t end)
{
    for (std::uint32_t i = begin; i < end; ++i)
    {
        Integrate(entities[i], dt);
    }
});
jobs.WaitIdle();
```

```
// Diagnostics: one row per thread, slot 0 is the main thread.
std::array<Fby::JobSystem::ThreadLoad, Fby::JobSystem::kMaxThreadSlots> loads{};
const std::uint32_t slots = jobs.SampleThreadLoad(loads);
```

ThreadLoad carries busyNanoseconds (wall time inside engine work), taskCount and a 16-byte name. The counters are one cache line per thread, so two workers finishing at the same time do not contend. SceneDiagnosticsPanel turns two samples into a per-thread occupancy percentage.

ThreadSlotCount() is ThreadCount() + 1 and is clamped by the fixed MCounters array; a device with more than 64 workers reports the first 64 slots.

6. ECS, scheduling and snapshots

The registry is FCS. Systems are registered on a SystemGraph in one of five phases (PreUpdate, FixedUpdate, Update, PostUpdate, Render). SystemContext is the only parameter a system receives; it carries the registry, JobSystem, frame arena, renderer, window, VFS, AssetDatabase, audio, physics, joints, cloth, fluid, camera stack, accessibility kernel and font atlases.

```
Fby::Ecs::Schedule schedule;
schedule.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem,
&physics);
schedule.Add(Fby::Ecs::Phase::Update, "movement", &MovementSystem, nullptr);
schedule.Add(Fby::Ecs::Phase::PostUpdate, "platform", &GogAchievementSystem,
&gog);

Fby::Ecs::SystemContext context{registry, jobs};
context.physics = &physics;
context.window = &window;
schedule.Run(Fby::Ecs::Phase::FixedUpdate, context);
```

Snapshot v5 is a canonical, checksummed byte image of the registry. Components are written by hand-written codecs when they contain runtime handles (b2BodyId, b2ShapeId); the snapshot never serializes those handles, and the restore path recreates bodies from the component fields. The schema stores the Mirror name and LayoutHash of every component, so a field change is rejected instead of silently misread.

API	Contract
GetOrAdd<T>(registry, entity)	Adds the component when missing and returns a stable reference.
ecs::SnapshotWriter / SnapshotReader	Writes and validates the canonical image; checksum is FNV-1a over the payload.
Fby::Net::NetRollbackSession	Consumes snapshot v5 and replays ticks on late input (section 10).

7. RHI, render items and frame statistics

The RHI is a hand-written Vulkan/Metal layer. Frames are recorded into persistent ring buffers; sprite items are 48-byte PODs; meshes are drawn through one indirect call per material/layer; linear-blend skinning is one compute dispatch. FrameStats exposes what the pass submitted, which is what SceneDiagnosticsPanel plots.

FrameStats field	Meaning
items / batches / drawCalls	Sprite items, batches and submitted draws.
meshItems / meshIndirectCalls	MeshRenderItem count and indirect draws.
skinDispatches / skinRanges	Skinning compute dispatches and ranges.
uploadedBytes	Bytes uploaded this frame.

The editor's Viewport renders into an offscreen target, not into the swapchain. EnsureViewportTarget() sizes that target in physical pixels; a target sized in points is the usual cause of a blurry Viewport on a HiDPI monitor.

8. Materials: cooked graphs, the pipeline cache and the shipped path

Material authoring and material compilation live on opposite sides of the engine boundary. `Render::MaterialGraph` is the authoring DAG: fixed capacities (128 nodes, 384 pins, 64 parameters), trivially copyable, and hashed without the editor positions. `Render::MaterialGraphCompiler` turns a validated graph into GLSL 450 text. `fby_matc` runs that compiler and glslang at build time and writes a `.femmat` payload, so a shipped game never links glslang and never parses GLSL. The full rationale is in [ADR 0002](../../adr/0002-materials-in-engine.md).

Piece	Header	Contract
`Render::MaterialGraph`	`Engine/RHI/MaterialGraph.hpp`	Authoring DAG; `Hash()` is the pipeline identity and excludes UI positions
`MaterialGraphCompiler`	`Engine/RHI/MaterialGraphCompiler.hpp`	DAG to GLSL 450 and the shader contract flags
`EncodeMaterialGraph` / `DecodeMaterialGraph`	`Engine/RHI/MaterialAsset.hpp`	`.fgraph` wire format: explicit little-endian fields, no struct memcpy
`EncodeCookedMaterial` / `DecodeCookedMaterial`	`Engine/RHI/MaterialAsset.hpp`	`.femmat` payload: 32-byte header, vertex and fragment SPIR-V, zero-copy decoding
`Render::MaterialPipelineCache`	`Engine/RHI/MaterialPipelineCache.hpp`	128 pipelines keyed by graph hash, reference counted, LRU eviction, no heap use
`Rhi::RHI::SupportsRuntimeShaderPipelines`	`Engine/RHI/RHI.hpp`	Capability query: Vulkan returns true, Null and Metal return false

```
# Author the graph once, cook it in the build step
build/bin/fby_matc assets/materials/tinted.fgraph -o
assets/materials/tinted.femmat
```

```
// `InBytes` is the .femmat payload from the VFS or a package entry.
Render::CookedMaterialView ScopedMaterial;
if (!Render::DecodeCookedMaterial(InBytes, &ScopedMaterial))
{
    return false;
}

// One cache per device: acquire at load time, release at shutdown.
const Rhi::ShaderPipelineHandle ScopedPipeline = InMaterials.Acquire(InRhi,
ScopedMaterial);

Rhi::MeshRenderItem ScopedItem;
ScopedItem.materialPipeline =
    ScopedPipeline.IsValid() ? ScopedPipeline.id :
Rhi::kDefaultMaterialPipeline;
```

- The graph hash is the pipeline identity. Editing a graph produces a new hash, so the old pipeline becomes an unreferenced cache entry instead of being mutated in place.
- The contract mask travels with the payload: generated vertex stage, fragment view projection, material push constants, world and vertex position varyings. A live entry whose hash arrives with another contract is refused, because that is a different shader.
- `Acquire()` adds a reference and `Release()` drops one. A pipeline at zero references stays resident, so a material acquired once per frame never recreates a pipeline; it is destroyed by LRU eviction, `ReleaseAll()` or `Shutdown()`.
- The cache never allocates and never formats text on the frame path; `LastError()` returns the fixed error buffer.
- Backends without runtime shader pipelines keep the engine shaders. Query `SupportsRuntimeShaderPipelines()` instead of probing for an invalid handle.

Payload	Fields	Validated on load
`.fgraph`	magic `FBYG`, version, node/pin/param counts, nodes, pins, params	magic, version, counts against capacity, exact size, node kinds, pin and param indices
`.femmat`	magic `FBYM`, version, contract mask, graph hash, vertex and fragment byte lengths	magic, version, size against the lengths, both SPIR-V modules, contract consistency

The editor keeps live editing on top of the same cache: `Fby::Editor::MaterialGraphRuntime` recompiles a graph when its hash changes, reports per-tab errors and delegates pipeline ownership to `MaterialPipelineCache`. `MaterialGraphRuntime::SetDomain` switches the live compile between the mesh and the sprite contract, and the panel's `SaveGraph`/`LoadGraph` write and read the

`.fgraph` payload the cooker consumes. A shipped build skips glslang entirely and only calls `DecodeCookedMaterial` and `Acquire`.

Sprites take materials through the third artefact: `MaterialCompileOptions::domain` selects the sprite domain, where the generated fragment stage matches the `quad.vert` varyings (`vUv`, `vColor`, `vLayer`, and `vWorld` when a node needs it) and samples the item's atlas slice through `vLayer`. `Time` is available: the sprite push block carries a material clock at offset 80, so the generated stage reads `pc.materialParams.x` exactly like a mesh shader. `fby_matc --sprite` cooks that variant. `DrawOrder::MaterialBucket` places the sprite in a bucket, `ChangeMaskRenderExtract::SetMaterialPipelines` maps buckets to cached pipeline ids, and the extractor emits one `DrawBatch` per bucket with `DrawBatch::pipeline` already resolved. The 64-bit render key carries the bucket (16 layer, 8 material, 32 depth, 8 sequence), so a material batch is never split by depth. Nodes that need the mesh view-projection (`Fresnel`, `VertexPosition`) are refused in the sprite domain with a readable reason.

9. Physics: Box2D v3, joints, cloth and SdfCollisionSystem

PhysicsSystem owns one b2WorldId and exposes it to extension modules as a packed uint32, so box2d.h never leaks into a public header. Bodies are mirrored from RigidBody2D/Collider2D components into Box2D, stepped at the fixed rate, and read back before interpolation. PhysicsJointSystem builds a fixed-capacity joint graph: chain, rope and spring connections with break limits, damage accumulation, gameplay cuts, repair, in-place rebuild and ties. ClothSystem integrates a Verlet grid and projects it against the same Box2D colliders.

SdfCollisionSystem contract:

Member	Contract
SetField(const SdfField&)	Installs a non-owning mdspan over a row-major grid. Negative values are solid; the field's memory is owned by the renderer/terrain, not by the system.
Step(registry, physics, fixedDt)	Samples the grid at body corners and centre, converts the gradient to a normal, applies a mass-scaled push impulse, reflects inward velocity and writes the corrected state back. Returns the number of corrected bodies.
Update(SystemContext&)	ECS convenience wrapper around Step(); a headless call without physics is a no-op.
Contacts()	Fixed array of SdfContact {entity, point, normal, penetration, impulse, sequence}.
SdfCollisionDesc	pushOut, restitution, maxImpulse, subSamples, dynamicOnly.

```
Fby::SdfCollisionDesc desc;
desc.pushOut = 0.65f;
desc.restitution = 0.05f;
desc.subSamples = 2;
```

```
sdf.Initialize(desc);
sdf.SetField(terrain.Field());
sdf.Register(graph);                    // Phase::FixedUpdate

// Viewport overlay: red vectors whose length is the penetration.
debug.Build(context, Fby::Editor::PhysicsDebugSdf, &sdf);
```

SdfCollisionSystem is not part of SystemContext. A host that wants SDF physics registers the system explicitly through Register(graph) and keeps the instance alive; the Viewport debug pass takes the same pointer.

10. NetRollbackSession: deterministic rollback

A session binds snapshot v5, CommandLog, Pcg32 and the gameplay components (tags, attributes, effects and skills) into one fixed-capacity rollback loop. Everything after Initialize() is allocation-free: one snapshot ring, one input ring per player, a fixed entity map and fixed frame metadata. The simulation callback is a function pointer plus void*.

Method	Contract
Initialize(registry, schema, config, arena)	Allocation-free setup with a caller-owned snapshot arena.
Initialize(registry, schema, config)	Convenience overload; allocates one arena (the only allocation the session makes).
SetSimulation(fn, user) / SetCommandLog(log)	Binds the pure simulation function and the structural command log.
Start(rng)	Captures the base tick and stores the session RNG. Required before Step().
Step(localInput)	Simulates exactly one tick with prediction for missing remote inputs and captures the snapshot.
SubmitRemoteInput(player, frame)	Stores a late or current input; a frame for an already simulated tick marks a pending rollback.
ReplayPending(report)	Restores the snapshot before the earliest corrected tick, restores RNG and CommandLog cursor, re-simulates, captures new snapshots and reports changed ticks.
CompareChecksums(tick, snap, state, out)	Compares a remote report with the stored frame; the two checksums isolate a byte-level desync from a gameplay-state desync.

```
Fby::Net::NetRollbackSession::Config cfg;
cfg.playerCount = 2;
cfg.localPlayer = 0;
cfg.rollbackTicks = 64;
session.Initialize(registry, schema, cfg, arena);
```

```
session.SetSimulation(&SimulateTick, &game);
session.Start(pcg);

while (running)
{
    session.SubmitRemoteInput(remote, received);
    session.Step(localInput);
    if (session.RollbackPending())
    {
        Fby::Net::RollbackReport report;
        session.ReplayPending(&report);
        if (report.changedTicks > 0)
        {
            LogBoy().Log(Fby::Editor::LogBoySeverity::Desync, "net",
                         "rollback {}..{} changed {}", report.fromTick,
report.toTick,
                         report.changedTicks);
        }
    }
}
```

- NetInputFrame is 24 bytes of integer axes (int16 fixed point). Floats are not allowed: a float bit pattern is not guaranteed to survive a different compiler.
- Result enumerates every failure mode (NotInitialized, BufferFull, RollbackTooOld, ChecksumMismatch, ...); ResultName() is the log spelling.
- A session is non-copyable and owns its arena when the convenience overload is used.

11. Audio: mixer, DSP graph, limiter

AudioSystem owns the miniaudio device and a fixed mixer. DspGraph routes nodes into buses, supports ducking and reverb, and receives commands through a lock-free ring so the game thread never blocks the audio callback. Limiter.hpp is the final true-peak stage.

- DeafMode mutes the graph at the Master root and turns sound triggers into subtitle lines; it is a policy of the audio system, not a post-effect.
- Ogg/Opus is decoded off the audio thread; the mixer only sees cooked .fpcm buffers.

12. PinkUI and WidgetCanvasSystem

PinkUI is the in-game UI: a fixed-capacity canvas, mdspan grid layout, anchor resolution, hit testing and one-batch packing. WidgetCanvasSystem turns a WidgetCanvas into FrameOutput (draw items) without allocating; font glyphs come from a .fbyfont atlas mapped through the VFS, so the shipping runtime carries no TTF rasterizer. Glyphs are atlas-only: there is no embedded 8x8 bitmap fallback, and a text frame without a mounted font array is refused instead of drawn with the wrong layer.

Member	Contract
WidgetCanvasSystem::MountFonts(RHI&)	Mandatory cold-start stage on the main thread. Builds the GPU texture array from the

	VFS .fbyfont set; a failure is a PANIC log plus FBY_ASSERT_MSG, never a fallback.
BuildFrame(context, canvas, output)	Lays out and culls one canvas, returning BuildStatus and filling FrameOutput. A text canvas without a current mounted atlas returns FontsNotMounted and writes nothing. Allocation-free; safe to call from Phase::Render.
Update(context, canvases, output)	Builds every active canvas into one output; the host submits the result as one batch. A font-contract violation stops the pass and sets fontContractViolated.
WidgetCanvas::SetDisplayScale	Density from CameraStackSystem::DisplayScale; font sizes are rounded to whole physical pixels.
OledShiftState	Integer 2 px orbit for OLED safety; ApplyOledShift() binds it to a canvas pool.

WidgetCanvasSystem is also the accessibility surface: BlindMode reads widget labels for the narrative log, DeafMode receives subtitle widgets, and colour-vision matrices plus FlashGuard are applied to the canvas colours before packing.

13. Accessibility kernel (BlindMode, DeafMode, FlashGuard)

Accessibility is a frame policy, not an overlay. BlindMode skips the game pass, indirect draws and the skinning dispatch, then publishes a bounded narrative log. DeafMode mutes the Master bus and routes sound triggers into SubtitleLines. FlashGuard compresses sudden luminance changes. Colour-vision correction uses the published Machado 2009 matrices, and SrgbRelativeLuminance()/ContrastRatio() implement WCAG 2.2 for authoring-time validation.

Item	Contract
NarrativeEvent	128-slot SPSC ring of 160-byte text payloads; the simulation thread pushes, the speech thread pops.
ColorVisionMatrix(deficiency)	Returns the 3x3 matrix for protanopia, deuteranopia or tritanopia.
ApplyColorVision(color, matrix, strength)	Applies the matrix in linear light to an sRGB Color32.
FlashGuard::Evaluate(luminance, dt)	Returns the gain for the current frame; a large delta is compressed to the threshold.
EditorSettings::ide*	The same policies applied to the editor window itself: Machado correction, WCAG high-contrast, FlashGuard, keyboard navigation and LogBoy as the narration sink.

14. Asset pipeline, VFS and localization

Assets are addressed by relative path inside the VFS. A .fempak archive is mounted with a priority; a directory mount can shadow it, which is how the editor writes a loose file before the cooker folds it back in. Read() returns a span into an mmap that stays valid until the mount is dropped or the file is hot-reloaded.

Format	Owner	Purpose
.fempak	FempakCooker	Immutable shipped archive with an index and optional encryption.
.fbyfont	FontAtlas (PinkUI)	Baked glyph atlas: [header] [glyph records][pixels]; no runtime TTF rasterizer.
.femeta	AssetDatabase	Typed metadata, import settings and stable AssetId.
.fbin	LocTool	Compiled localization table; the editor UI itself is English-only.

The editor UI is not localizable by design: menus, panel titles, errors and the inspector are English-only. Localization applies to game content and documentation, where a .fbin table and the EN/RU manual cover the shipped languages.

15. Platform services: Steam, Epic Games Store, GOG

PlatformService is the engine-side bridge; PlatformAPI.h is a flat pure-C ABI that a store SDK plugin implements. The provider is loaded dynamically, resolves only documented entry points and converts every store-specific type at the boundary, so a game never includes steam_api.h, EOSSDK or Galaxy headers.

Provider	Status in this tree	Notes
SteamProvider.cpp	Implemented	Steamworks flat API through a function table.
EgsProvider.cpp	Implemented	Epic Online Services provider through the same C ABI.
GogProvider.cpp	Implemented (shim)	Loads galaxy64.dll / libgalaxy64.so / libgalaxy64.dylib at runtime and resolves the flat fby_gog_* shim ABI; the game ships the shim built against its SDK revision. Galaxy is a C++ interface library, so a shim is mandatory. GogAchievementSystem and GogStorageSystem write the

		binary PlatformCommandLog in Phase::PostUpdate.
fby_fontc	Implemented	Offline TTF -> .fbyfont cooker (stb_truetype, RGBA8 atlas, verified header).

```
// Pure C ABI: the provider is a shared library, not a C++ dependency.
typedef struct FbyPlatformApi {
    uint32_t version;
    void* user;
    bool (*init)(void* user);
    void (*shutdown)(void* user);
    void (*unlockAchievement)(void* user, const char* id);
    bool (*writeCloudFile)(void* user, const char* name, const uint8_t* data,
uint32_t size);
} FbyPlatformApi;
```

16. Editor: dashboard, fonts, LogBoy, Scene Diagnostics

The dashboard is the only editor surface that exists before a project is open. It installs the editor font set before the first frame, so the first window is already Inter at the physical DPI. A new project starts in Projects/ under the engine root, resolved by DefaultProjectsRoot() - never in the process working directory and never in the install directory.

Component	Contract
EditorFonts	Loads Inter and JetBrains Mono from the VFS mmap. Subpixel advances (PixelSnapH=false), auto oversampling, no bitmap fallback, dynamic atlas baked at the viewport pixel density. Legacy backends get RasterizerDensity + inverse FontGlobalScale.
ProjectDashboard	Initialize(paths, config, visible, window, vfs, fonts, uiScale): installs typography, resolves the default Projects/ location and creates it.
LogBoySystem / LogBoyPanel	Fixed 4096-record ring of 192-byte text payloads with severity Trace/Info/Warning/Critical/Desync, FNV-1a system hashes and colour+hash filtering.
SceneDiagnosticsPanel	Per-thread occupancy bars and heatmap, FrameArena bytes, MeshRenderItem count, indirect draws, draw calls and exact simulation microseconds with avg/p99/peak.
PhysicsDebugVisualizer	Fixed-buffer line/point geometry for Box2D bodies, ropes, cloth and SDF normals, drawn over the Viewport through a caller-supplied world-to-screen map.

The .fbyfont atlas is a baked PinkUI asset and cannot feed ImGui's vector rasterizer. The editor therefore loads the Inter TTF from the same VFS mount; the atlas header is still validated on startup, so a corrupt game-UI atlas is reported by the editor, not by the player.

17. Step-by-step: a game from an empty folder

This walkthrough produces a playable project with a sprite, physics, keyboard input, a PinkUI HUD and a deterministic rollback session. Every step maps to a real header from the index.

- 1. Launch the editor and create a project with New Project. The dashboard creates <engineRoot>/Projects/<name> with assets/{textures,audio,scenes}, configs/ and project.femproj.
- 2. Import a sprite into assets/textures. The watcher imports it, writes a .femeta and cooks a .ftex texture array layer.
- 3. Create assets/scenes/main.toml with one SpriteRenderer, one Transform2D and one RigidBody2D + Collider2D.
- 4. Register the systems: input in PreUpdate, physics in FixedUpdate, movement in Update, PinkUI in Render.
- 5. Give the camera a CameraStackSystem and call SetDisplayScale() from SDL_GetWindowDisplayScale so UI glyphs are crisp on HiDPI.
- 6. Add an AccessibilityKernel from project settings; BlindMode and DeafMode are read from project.femproj, not hard-coded.
- 7. Create a NetRollbackSession per match, bind a pure SimulateTick callback and feed NetInputFrame values.
- 8. Log through LogBoy: severity + system hash + fixed-format message. Open the LogBoy panel and filter by Desync while testing netcode.
- 9. Enable Physics Debug Draw in the Viewport while tuning colliders and cloth.
- 10. Cook with FempakCooker, run the suite with ctest and the style gate, then ship the engine-only target.

```
// 4-7 in code: systems, accessibility and rollback for one match.
Fby::Ecs::Schedule schedule;
schedule.Add(Fby::Ecs::Phase::PreUpdate, "input", &InputSystem);
schedule.Add(Fby::Ecs::Phase::FixedUpdate, "physics", &PhysicsStepSystem,
&physics);
schedule.Add(Fby::Ecs::Phase::Update, "movement", &MovementSystem);
schedule.Add(Fby::Ecs::Phase::Render, "ui", &UiSystem, &ui);

Fby::AccessibilityKernel kernel;
kernel.Configure(settings);              // blind/deaf/colour/flash

Fby::Net::NetRollbackSession session;
if (session.Initialize(registry, schema, cfg, arena) !=
Fby::Net::NetRollbackSession::Result::Ok)
{
    LogBoy().Log(Fby::Editor::LogBoySeverity::Critical, "net", "rollback init
failed");
```

```
}
```

18. Testing, style gate and release checklist

The suite uses no test framework: every test is an executable with a main() that returns non-zero on failure, so it runs on a headless CI agent, a console devkit and a developer workstation with the same command. The full suite passes at Release.

```
ctest --test-dir build --output-on-failure          # the full suite
python Source/Tools/Dev/CheckStyle.py               # style gate
python Documentation/Tools/GenerateAPIReference.py  # this manual
python Documentation/Tools/GenerateDocumentation.py
python Documentation/Tools/GenerateApplicationGuide.py
```

- A release build enables asserts and runs the whole suite once with FBY_ENABLE_ASSERTS=1, because two latent debug-only compile errors existed in JobSystem at RC.
- The POSIX branch of WriteFileAtomically() and LogLine() must be compiled on Linux/macOS in CI; the Windows-only build never saw them.
- The style gate must report zero failures before a commit is eligible for release.

19. Open work

Open work is tracked in Documentation/ROADMAP.md.

20. Appendix A - generated symbol index

Generated from the headers on every run. Header and line are the authoritative location.

Editor/Infrastructure (119)

Symbol	Kind	Header	Line
AccessibilityEffectRegistry	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	47
AIStateTreeCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	54
AnimationGraphCanvasPanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	55
AnimationTimelinePanel	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	56
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	20
AssetDatabaseWatchSystem	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	57
AssetImportService	Infrastructure	Source/Editor/Infrastructure/AssetImportService.hpp	30
AssetImportService	Infrastructure	Source/Editor/Infrastructure/EditorSystem.hpp	58
Callbacks	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	38
Change	Infrastructure	Source/Editor/Infrastructure/AssetDatabaseWatchSystem.hpp	126
Codec	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	222
Command	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	236

CommandKind	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	33
CommandStack	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	30
CommandStack	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	59
Config	Infrastructure	Source/Editor/ Infrastructure/ AssetDatabaseWatch System.hpp	27
Config	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	39
Config	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	88
ContentBrowserPan el	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	60
CookRequest	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	141
CookResult	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	152
EditorCamera	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	101
EditorDesc	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	72
EditorFontRole	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	22
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	84
EditorFonts	Infrastructure	Source/Editor/ Infrastructure/	7

		EditorTypography.hpp	
EditorFontSpec	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	38
EditorFontStatus	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	67
EditorPalette	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	77
EditorSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorSettings.hpp	20
EditorSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	14
EditorSettingsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	61
EditorSymbolColors	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	61
EditorSymbolRole	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	42
EditorSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	110
EditorThemeId	Infrastructure	Source/Editor/ Infrastructure/ EditorTheme.hpp	31
Entry	Infrastructure	Source/Editor/ Infrastructure/ TextureThumbnails.hp p	52
ExternalClientDesc	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	37
FrameGenerationMo de	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	139
FrameGenerationPol icySetting	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	132
HudTargetMode	Infrastructure	Source/Editor/	148

		Infrastructure/ ProjectConfig.hpp	
ImFont	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	10
ImGuiIO	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	11
Impl	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	241
InteractionTransaction	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	16
LoadedSource	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	138
MaterialGraphPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	62
MaterialGraphRuntime	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime .hpp	41
MaterialRuntimeStats	Infrastructure	Source/Editor/ Infrastructure/ MaterialGraphRuntime .hpp	25
MotionComfortSettings	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	171
NetSession	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	32
NetworkDiagnosticsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	63
OitMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	99
OpaqueRenderMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	92
PendingEdit	Infrastructure	Source/Editor/ Infrastructure/	261

		CommandStack.hpp	
PendingSpawn	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	270
PhysicsDebugFlags	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	19
PhysicsDebugFrame	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	47
PhysicsDebugLine	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	32
PhysicsDebugPoint	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	40
PhysicsDebugVisualizer	Infrastructure	Source/Editor/ Infrastructure/ PhysicsDebugDraw.hpp	58
PiePlayerInput	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	27
PlaybackIcon	Infrastructure	Source/Editor/ Infrastructure/ EditorWidgets.hpp	12
PlayInEditor	Infrastructure	Source/Editor/ Infrastructure/ PlayInEditor.hpp	44
PluginHost	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	42
Profiler	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	27
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	67
ProjectConfig	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	286

ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	64
ProjectConfigStore	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	350
ProjectDashboard	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	65
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	68
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	308
ProjectPaths	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	21
ProjectPluginEntry	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	163
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	182
ProjectSettings	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	22
ProjectSettingsPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	66
Range	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	253
ReleaseCookConfig	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	70
ReleaseCooker	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	158
ReleaseCookProgress	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	121
ReleaseCookResult	Infrastructure	Source/Editor/ Infrastructure/ ReleaseCooker.hpp	142

ReleaseCookStage	Infrastructure	Source/Editor/Infrastructure/ReleaseCooker.hpp	28
ResolvedHudSettings	Infrastructure	Source/Editor/Infrastructure/ResolvedRenderSettings.hpp	35
ResolvedRenderSettings	Infrastructure	Source/Editor/Infrastructure/ResolvedRenderSettings.hpp	14
Result	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	43
Result	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	47
SdfCollisionSystem	Infrastructure	Source/Editor/Infrastructure/PhysicsDebugDraw.hpp	13
Services	Infrastructure	Source/Editor/Infrastructure/PlayInEditor.hpp	224
SettingDesc	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	28
SettingKind	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	20
SettingScope	Infrastructure	Source/Editor/Infrastructure/SettingsSchema.hpp	11
Slot	Infrastructure	Source/Editor/Infrastructure/MaterialGraphRuntime.hpp	118
Source	Infrastructure	Source/Editor/Infrastructure/EditTransactions.hpp	19
SpawnComponent	Infrastructure	Source/Editor/Infrastructure/CommandStack.hpp	284
SpawnHeader	Infrastructure	Source/Editor/Infrastructure/	277

		CommandStack.hpp	
SpawnTransaction	Infrastructure	Source/Editor/ Infrastructure/ EditTransactions.hpp	190
StableSource	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	145
Stats	Infrastructure	Source/Editor/ Infrastructure/ AssetDatabaseWatch System.hpp	88
Stats	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	55
Stats	Infrastructure	Source/Editor/ Infrastructure/ CommandStack.hpp	95
TemporalUpscaleMo de	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	155
TextureArrayAsset	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	23
TextureBinding	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	168
TextureThumbnailCa che	Infrastructure	Source/Editor/ Infrastructure/ TextureThumbnails.hp p	17
UpscaleMode	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	106
UpscalePolicy	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	115
UpscalerBackendSet ting	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	122
UserSettings	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	69

UserSettings	Infrastructure	Source/Editor/ Infrastructure/ UserSettings.hpp	16
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ AssetImportService.hp p	20
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	15
VirtualFileSystem	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	18
VirtualTextureSettin gs	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	37
VirtualTextureSettin gs	Infrastructure	Source/Editor/ Infrastructure/ ProjectConfig.hpp	32
VirtualTextureSyste m	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	22
VisualScriptPanel	Infrastructure	Source/Editor/ Infrastructure/ EditorSystem.hpp	70
WindowManager	Infrastructure	Source/Editor/ Infrastructure/ EditorFonts.hpp	16

Editor/Panels (90)

Symbol	Kind	Header	Line
Action	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	25
AIStateTreeCanvasP anel	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	55
AIStateTreeCompile d	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	35
AIStateTreeDocume nt	Panels	Source/Editor/ Panels/	24

		AIStateTreeCanvasPanel.hpp	
AnimationGraphCanvasPanel	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	125
AnimationGraphDocument	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	88
AnimationGraphLink	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	74
AnimationGraphLinkKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	37
AnimationGraphNode	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	49
AnimationGraphNodeKind	Panels	Source/Editor/Panels/AnimationGraphCanvasPanel.hpp	29
AnimationTimelinePanel	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	34
AssetImportService	Panels	Source/Editor/Panels/ContentBrowserPanel.hpp	24
AudioSystem	Panels	Source/Editor/Panels/AnimationTimelinePanel.hpp	15
AudioSystem	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	11
CanvasModelOps	Panels	Source/Editor/Panels/NodeCanvasPanel.hpp	21
CommandStack	Panels	Source/Editor/	25

		Panels/ ContentBrowserPanel. hpp	
CommandStack	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	30
ComponentRegistry	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel. hpp	18
Config	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	35
ContentBrowserPan el	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	27
DirectionalBlendSpa cePanel	Panels	Source/Editor/ Panels/ DirectionalBlendSpac ePanel.hpp	24
DragPayload	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	60
EditorFonts	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	20
EditorFonts	Panels	Source/Editor/ Panels/ VisualScriptPanel.hpp	20
EditorSettingsPanel	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	91
EditSource	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	58
EditSource	Panels	Source/Editor/ Panels/ AnimationGraphCanv asPanel.hpp	128
Entry	Panels	Source/Editor/ Panels/ ContentBrowserPanel.	43

		hpp	
FileDialog	Panels	Source/Editor/ Panels/ ContentBrowserPanel. hpp	140
ImNodesContext	Panels	Source/Editor/ Panels/ AIStateTreeCanvasPa nel.hpp	13
ImNodesContext	Panels	Source/Editor/ Panels/ AnimationGraphCanv asPanel.hpp	15
LevelSettingsPanel	Panels	Source/Editor/ Panels/ LevelSettingsPanel.hp p	12
LicensesPanel	Panels	Source/Editor/ Panels/ LicensesPanel.hpp	11
LogBoyPanel	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	394
LogBoyRecord	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	105
LogBoyRing	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	121
LogBoySeverity	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	33
LogBoySystem	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	186
LogicGraphDocume nt	Panels	Source/Editor/ Panels/ VisualScriptPanel.hpp	23
MaterialGraphPanel	Panels	Source/Editor/ Panels/ MaterialGraphPanel.h pp	20
Mode	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	84

NetSession	Panels	Source/Editor/ Panels/ NetworkDiagnosticsPanel.hpp	6
NetworkDiagnosticsPanel	Panels	Source/Editor/ Panels/ NetworkDiagnosticsPanel.hpp	15
NodeCanvasPanel	Panels	Source/Editor/ Panels/ NodeCanvasPanel.hpp	54
ParticleFxPanel	Panels	Source/Editor/ Panels/ ParticleFxPanel.hpp	16
PIEAnimFrame	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	110
PieClientViewport	Panels	Source/Editor/ Panels/ PieClientViewport.hpp	15
PIEFrameSlot	Panels	Source/Editor/ Panels/ AnimationGraphCanvasPanel.hpp	302
PlaneView	Panels	Source/Editor/ Panels/ DirectionalBlendSpacePanel.hpp	122
ProjectConfigStore	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	31
ProjectConfigStore	Panels	Source/Editor/ Panels/ VoiceDevicePanel.hpp	24
ProjectDashboard	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	22
ProjectSettingsPanel	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	33
QueryKind	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	430
SceneDiagnosticsPa	Panels	Source/Editor/	56

nel		Panels/ SceneDiagnosticsPanel.hpp	
SceneDiagnosticsSample	Panels	Source/Editor/ Panels/ SceneDiagnosticsPanel.hpp	15
ScriptDebuggerPanel	Panels	Source/Editor/ Panels/ ScriptDebuggerPanel.hpp	28
ScriptRuntime	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	16
SdfCharacterPanel	Panels	Source/Editor/ Panels/ SdfCharacterPanel.hpp	9
SinkUser	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	88
SkeletonInspectorState	Panels	Source/Editor/ Panels/ SkeletonAssetPanel.hpp	19
SpawnContext	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	68
SpriteAnimationInspectorState	Panels	Source/Editor/ Panels/ SpriteAnimationPanel.hpp	21
SystemContext	Panels	Source/Editor/ Panels/ SettingsPanels.hpp	24
SystemContext	Panels	Source/Editor/ Panels/ SkeletonAssetPanel.hpp	13
SystemContext	Panels	Source/Editor/ Panels/ SpriteAnimationPanel.hpp	13

SystemName	Panels	Source/Editor/ Panels/ LogBoyPanel.hpp	189
TextPreviewPanel	Panels	Source/Editor/ Panels/ TextPreviewPanel.hpp	24
TextureThumbnailCache	Panels	Source/Editor/ Panels/ SpriteAnimationPanel.hpp	19
TilemapEditorPanel	Panels	Source/Editor/ Panels/ TilemapEditorPanel.hpp	20
TimelinePanelContext	Panels	Source/Editor/ Panels/ AnimationTimelinePanel.hpp	22
TransportGroupPanel	Panels	Source/Editor/ Panels/ TransportGroupPanel.hpp	14
UserSettingsPanel	Panels	Source/Editor/ Panels/ UserSettingsPanel.hpp	16
VirtualFileSystem	Panels	Source/Editor/ Panels/ ContentBrowserPanel.hpp	18
VirtualFileSystem	Panels	Source/Editor/ Panels/ ProjectDashboard.hpp	13
VirtualFileSystem	Panels	Source/Editor/ Panels/ TextPreviewPanel.hpp	15
VirtualTexturePanel	Panels	Source/Editor/ Panels/ VirtualTexturePanel.hpp	18
VirtualTextureSystem	Panels	Source/Editor/ Panels/ VirtualTexturePanel.hpp	10
VisualScriptPanel	Panels	Source/Editor/ Panels/	24

		ScriptDebuggerPanel.hpp	
VisualScriptPanel	Panels	Source/Editor/Panels/VisualScriptPanel.hpp	32
VoiceDevicePanel	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	26
VoiceEffectPolicy	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	17
VoicePresetChoice	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	15
VoicePresetRegistry	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	14
VoicePresetSelection	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	16
VoiceReactiveSfxSettings	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	18
VoiceSession	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	12
VoiceSystem	Panels	Source/Editor/Panels/VoiceDevicePanel.hpp	13
WidgetCanvas	Panels	Source/Editor/Panels/SettingsPanels.hpp	18
WindowManager	Panels	Source/Editor/Panels/ProjectDashboard.hpp	14

Engine/AI (7)

Symbol	Kind	Header	Line
AIAgentSettings	AI	Source/Engine/AI/AIController.hpp	17
AIControllerComponent	AI	Source/Engine/AI/AIController.hpp	32
AIControllerSystem	AI	Source/Engine/AI/	96

		AIController.hpp	
AIPerceptionComponent	AI	Source/Engine/AI/AIController.hpp	60
AIPerceptionSettings	AI	Source/Engine/AI/AIController.hpp	49
AIPerceptionSystem	AI	Source/Engine/AI/AIController.hpp	68
GridEntry	AI	Source/Engine/AI/AIController.hpp	83

Engine/Assets (64)

Symbol	Kind	Header	Line
AeadAlgorithm	Assets	Source/Engine/Assets/FemCrypto.hpp	18
AeadKey	Assets	Source/Engine/Assets/FemCrypto.hpp	39
AeadNonce	Assets	Source/Engine/Assets/FemCrypto.hpp	46
AeadTag	Assets	Source/Engine/Assets/FemCrypto.hpp	54
ArchiveBuildOptions	Assets	Source/Engine/Assets/FemArchiver.hpp	102
ArchiveCipher	Assets	Source/Engine/Assets/FemArchiver.hpp	29
ArchiveCodec	Assets	Source/Engine/Assets/FemArchiver.hpp	22
ArchiveEntryView	Assets	Source/Engine/Assets/FemArchiver.hpp	118
ArchiveInput	Assets	Source/Engine/Assets/FemArchiver.hpp	72
ArchiveProgress	Assets	Source/Engine/Assets/FemArchiver.hpp	85
ArchiveProgressSta	Assets	Source/Engine/	78

ge		Assets/ FemArchiver.hpp	
ArchiveReader	Assets	Source/Engine/ Assets/ FemArchiver.hpp	129
ArchiveReader	Assets	Source/Engine/ Assets/ FemArchiver.hpp	148
ArchiveStatus	Assets	Source/Engine/ Assets/ FemArchiver.hpp	36
AssetDatabase	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	36
AssetDependency	Assets	Source/Engine/ Assets/AssetMeta.hpp	83
AssetHandle	Assets	Source/Engine/ Assets/AssetMeta.hpp	22
AssetImportSettings	Assets	Source/Engine/ Assets/AssetMeta.hpp	45
AssetMeta	Assets	Source/Engine/ Assets/AssetMeta.hpp	61
AssetRecord	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	27
AstcBlock	Assets	Source/Engine/ Assets/ TextureCook.hpp	25
AtlasDesc	Assets	Source/Engine/ Assets/ TextureArray.hpp	31
AtlasLayer	Assets	Source/Engine/ Assets/ TextureArray.hpp	21
AtlasPackError	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	57
AtlasSprite	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	26
AtlasSpriteSource	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	18
AtlasUv	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	41
AudioClipHeader	Assets	Source/Engine/ Assets/	87

		AudioClipFormat.hpp	
BlockExtent	Assets	Source/Engine/ Assets/ TextureFormat.hpp	103
CleanRenderExtract	Assets	Source/Engine/ Assets/ ProjectScaffold.hpp	133
CookedAudioView	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	116
CookedTexture	Assets	Source/Engine/ Assets/ TextureCook.hpp	41
CookedTextureView	Assets	Source/Engine/ Assets/ TextureFormat.hpp	203
CookProgressFlags	Assets	Source/Engine/ Assets/ CookProgress.hpp	31
CookProgressRecord	Assets	Source/Engine/ Assets/ CookProgress.hpp	38
CookProgressStage	Assets	Source/Engine/ Assets/ CookProgress.hpp	18
CookTarget	Assets	Source/Engine/ Assets/ TextureCook.hpp	17
FanmHeader	Assets	Source/Engine/ Assets/ AnimationClipAsset.hpp	22
FemArchiver	Assets	Source/Engine/ Assets/ FemArchiver.hpp	131
FempakInput	Assets	Source/Engine/ Assets/ FempakWriter.hpp	15
FempakWriter	Assets	Source/Engine/ Assets/ FempakWriter.hpp	22
LocaleTable	Assets	Source/Engine/ Assets/ Localization.hpp	14
LocalizationGender	Assets	Source/Engine/	21

		Assets/ Localization.hpp	
LocalizationPack	Assets	Source/Engine/ Assets/ Localization.hpp	13
LocalizationTable	Assets	Source/Engine/ Assets/ Localization.hpp	28
MappedFile	Assets	Source/Engine/ Assets/VFS.hpp	20
MountRecord	Assets	Source/Engine/ Assets/VFS.hpp	134
PakEntry	Assets	Source/Engine/ Assets/ PakFormat.hpp	26
PakHeader	Assets	Source/Engine/ Assets/ PakFormat.hpp	16
PcmFormat	Assets	Source/Engine/ Assets/ AudioClipFormat.hpp	21
PinnedFile	Assets	Source/Engine/ Assets/VFS.hpp	67
ReloadNotice	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	94
SpriteAtlas	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	47
SpriteAtlasAsset	Assets	Source/Engine/ Assets/SpriteAtlas.hpp	77
StagedRow	Assets	Source/Engine/ Assets/ AssetDatabase.hpp	116
StringEntry	Assets	Source/Engine/ Assets/ Localization.hpp	15
TextureAlphaMode	Assets	Source/Engine/ Assets/ TextureFormat.hpp	43
TextureArrayAsset	Assets	Source/Engine/ Assets/ TextureArray.hpp	51
TextureCookOptions	Assets	Source/Engine/ Assets/	31

		TextureCook.hpp	
TextureFormat	Assets	Source/Engine/ Assets/ TextureFormat.hpp	22
TextureHeader	Assets	Source/Engine/ Assets/ TextureFormat.hpp	184
VfsEntry	Assets	Source/Engine/ Assets/VFS.hpp	59
VirtualFileSystem	Assets	Source/Engine/ Assets/ TextureArray.hpp	16
VirtualFileSystem	Assets	Source/Engine/ Assets/VFS.hpp	73

Engine/Audio (122)

Symbol	Kind	Header	Line
Applied	Audio	Source/Engine/ Audio/ VoicePreset.hpp	154
AudioBus	Audio	Source/Engine/ Audio/Audio.hpp	27
AudioBusComponent	Audio	Source/Engine/ Audio/Audio.hpp	44
AudioCaptureMonitor	Audio	Source/Engine/ Audio/ AudioDevices.hpp	144
AudioClip	Audio	Source/Engine/ Audio/Audio.hpp	75
AudioCommand	Audio	Source/Engine/ Audio/DspGraph.hpp	138
AudioCommandType	Audio	Source/Engine/ Audio/DspGraph.hpp	121
AudioContextComponent	Audio	Source/Engine/ Audio/ AudioZones.hpp	69
AudioDeviceEnumerator	Audio	Source/Engine/ Audio/ AudioDevices.hpp	93
AudioDeviceInfo	Audio	Source/Engine/ Audio/ AudioDevices.hpp	54

AudioDeviceKind	Audio	Source/Engine/ Audio/ AudioDevices.hpp	15
AudioDeviceList	Audio	Source/Engine/ Audio/ AudioDevices.hpp	66
AudioDevicePreferences	Audio	Source/Engine/ Audio/ AudioDevices.hpp	73
AudioDeviceStatus	Audio	Source/Engine/ Audio/ AudioDevices.hpp	21
AudioLimiter	Audio	Source/Engine/ Audio/Limiter.hpp	23
AudioListener	Audio	Source/Engine/ Audio/Audio.hpp	100
AudioSource	Audio	Source/Engine/ Audio/Audio.hpp	85
AudioSourceComponent	Audio	Source/Engine/ Audio/Audio.hpp	57
AudioSystem	Audio	Source/Engine/ Audio/Audio.hpp	106
AudioSystem	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	15
AudioZoneComponent	Audio	Source/Engine/ Audio/ AudioZones.hpp	34
AudioZoneKind	Audio	Source/Engine/ Audio/ AudioZones.hpp	14
AudioZoneSystem	Audio	Source/Engine/ Audio/ AudioZones.hpp	93
BiquadState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	369
BusAtomics	Audio	Source/Engine/ Audio/DspGraph.hpp	370
BusFilterParams	Audio	Source/Engine/ Audio/DspGraph.hpp	108
BusRuntime	Audio	Source/Engine/ Audio/Audio.hpp	227
BusShadow	Audio	Source/Engine/	291

		Audio/DspGraph.hpp	
BusState	Audio	Source/Engine/ Audio/DspGraph.hpp	338
CommandQueue	Audio	Source/Engine/ Audio/DspGraph.hpp	170
CookedClipRecord	Audio	Source/Engine/ Audio/Audio.hpp	202
DspClipView	Audio	Source/Engine/ Audio/DspGraph.hpp	205
DspGraph	Audio	Source/Engine/ Audio/DspGraph.hpp	216
DspGraph	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	16
DspResult	Audio	Source/Engine/ Audio/DspGraph.hpp	34
DspResult	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	19
DuckingParams	Audio	Source/Engine/ Audio/DspGraph.hpp	80
DuckingState	Audio	Source/Engine/ Audio/DspGraph.hpp	298
Entry	Audio	Source/Engine/ Audio/ VoicePreset.hpp	83
FilterState	Audio	Source/Engine/ Audio/DspGraph.hpp	327
FloatRing	Audio	Source/Engine/ Audio/Audio.hpp	25
FloatRing	Audio	Source/Engine/ Audio/ AudioStream.hpp	16
FloatRing	Audio	Source/Engine/ Audio/DspGraph.hpp	32
JitterSlot	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	119
LimiterDesc	Audio	Source/Engine/ Audio/Limiter.hpp	11
OggOpusStream	Audio	Source/Engine/ Audio/OggOpus.hpp	131

OggPacket	Audio	Source/Engine/ Audio/OggOpus.hpp	76
OggPacketReader	Audio	Source/Engine/ Audio/OggOpus.hpp	86
OggPage	Audio	Source/Engine/ Audio/OggOpus.hpp	36
OggPageReader	Audio	Source/Engine/ Audio/OggOpus.hpp	57
OpusHeadInfo	Audio	Source/Engine/ Audio/OggOpus.hpp	106
OwnedPcmRecord	Audio	Source/Engine/ Audio/Audio.hpp	209
Peer	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	131
PlayerState	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	88
RateBucket	Audio	Source/Engine/ Audio/ VoiceNetworkBridge.hpp	62
ResolvedZone	Audio	Source/Engine/ Audio/ AudioZones.hpp	115
ReverbParams	Audio	Source/Engine/ Audio/DspGraph.hpp	96
ReverbState	Audio	Source/Engine/ Audio/DspGraph.hpp	311
SlotRuntime	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	385
SourceRuntime	Audio	Source/Engine/ Audio/Audio.hpp	219
State	Audio	Source/Engine/ Audio/ AudioDevices.hpp	164
Stats	Audio	Source/Engine/ Audio/ AudioZones.hpp	96
VoiceAec	Audio	Source/Engine/ Audio/VoiceAec.hpp	44
VoiceAecSettings	Audio	Source/Engine/	22

		Audio/VoiceAec.hpp	
VoiceCapture	Audio	Source/Engine/ Audio/ AudioStream.hpp	64
VoiceChannel	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	65
VoiceChannelConfig	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	36
VoiceChannelStats	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	46
VoiceCodecConfig	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	15
VoiceConditionOutput	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	333
VoiceConditionSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	284
VoiceConditionState	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	324
VoiceDecoder	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	51
VoiceDenoiseTier	Audio	Source/Engine/ Audio/VoiceDsp.hpp	10
VoiceDistortionSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	157
VoiceDspChain	Audio	Source/Engine/ Audio/VoiceDsp.hpp	99
VoiceDspSettings	Audio	Source/Engine/ Audio/VoiceDsp.hpp	43
VoiceEffectChain	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	345
VoiceEffectChainSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	221
VoiceEffectKind	Audio	Source/Engine/	20

		Audio/ VoiceEffects.hpp	
VoiceEffectPolicy	Audio	Source/Engine/ Audio/ VoicePreset.hpp	116
VoiceEffectPresetKind	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	235
VoiceEffectProfile	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	310
VoiceEffectRack	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	433
VoiceEffectSlot	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	200
VoiceEncoder	Audio	Source/Engine/ Audio/ VoiceCodec.hpp	27
VoiceFormantSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	146
VoiceMuffleSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	80
VoiceMuteSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	67
VoiceNetworkBridge	Audio	Source/Engine/ Audio/ VoiceNetworkBridge.hpp	18
VoiceParticipant	Audio	Source/Engine/ Audio/ VoiceSession.hpp	46
VoicePitchSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	133
VoicePresetApplier	Audio	Source/Engine/ Audio/ VoicePreset.hpp	146
VoicePresetCategory	Audio	Source/Engine/ Audio/ VoicePreset.hpp	22

VoicePresetChoice	Audio	Source/Engine/ Audio/ VoicePreset.hpp	95
VoicePresetDesc	Audio	Source/Engine/ Audio/ VoicePreset.hpp	47
VoicePresetFlags	Audio	Source/Engine/ Audio/ VoicePreset.hpp	35
VoicePresetKind	Audio	Source/Engine/ Audio/VoiceDsp.hpp	68
VoicePresetRegistry	Audio	Source/Engine/ Audio/ VoicePreset.hpp	58
VoicePresetResolution	Audio	Source/Engine/ Audio/ VoicePreset.hpp	124
VoicePresetSelection	Audio	Source/Engine/ Audio/ VoicePreset.hpp	105
VoiceRadioSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	115
VoiceReactiveSfx	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	79
VoiceReactiveSfxInput	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	53
VoiceReactiveSfxPlayer	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	43
VoiceReactiveSfxSettings	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	58
VoiceRingModulationSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	171
VoiceRouting	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	30
VoiceSession	Audio	Source/Engine/ Audio/ VoiceChannel.hpp	15
VoiceSession	Audio	Source/Engine/	15

		Audio/ VoicePreset.hpp	
VoiceSession	Audio	Source/Engine/ Audio/ VoiceSession.hpp	65
VoiceSfxEvent	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	31
VoiceSfxKind	Audio	Source/Engine/ Audio/ VoiceReactiveSfx.hpp	17
VoiceSpatialPosition	Audio	Source/Engine/ Audio/ VoiceSession.hpp	37
VoiceSpatialSettings	Audio	Source/Engine/ Audio/ VoiceSession.hpp	22
VoiceState	Audio	Source/Engine/ Audio/DspGraph.hpp	350
VoiceSystem	Audio	Source/Engine/ Audio/ VoicePreset.hpp	16
VoiceSystem	Audio	Source/Engine/ Audio/ VoiceSystem.hpp	39
VoiceSystemConfig	Audio	Source/Engine/ Audio/ VoiceSystem.hpp	22
VoiceUnderwaterSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	98
VoiceWhisperSettings	Audio	Source/Engine/ Audio/ VoiceEffects.hpp	185
VorbisIdentInfo	Audio	Source/Engine/ Audio/OggOpus.hpp	121

Engine/Core (113)

Symbol	Kind	Header	Line
AssetId	Core	Source/Engine/Core/ AssetId.hpp	43
AssetKind	Core	Source/Engine/Core/ AssetId.hpp	15

AudioBus	Core	Source/Engine/Core/Pause.hpp	11
AudioSystem	Core	Source/Engine/Core/Pause.hpp	10
BigIntT	Core	Source/Engine/Core/BigInt.hpp	179
BigNumberStore	Core	Source/Engine/Core/BigNumberStore.hpp	41
BinaryKind	Core	Source/Engine/Core/BigNumberStore.hpp	259
BinaryKind	Core	Source/Engine/Core/DecimalStore.hpp	206
BufferTag	Core	Source/Engine/Core/Types.hpp	301
BuildInfo	Core	Source/Engine/Core/BuildInfo.hpp	12
BuildProfile	Core	Source/Engine/Core/BuildProfile.hpp	135
CameraView	Core	Source/Engine/Core/Grid.hpp	337
Cell	Core	Source/Engine/Core/Grid.hpp	74
CivilDate	Core	Source/Engine/Core/Calendar.hpp	23
Clock	Core	Source/Engine/Core/Time.hpp	28
ClockTime	Core	Source/Engine/Core/Calendar.hpp	34
Color32	Core	Source/Engine/Core/Types.hpp	203
ComponentDesc	Core	Source/Engine/Core/Mirror.hpp	51
ComponentRegistry	Core	Source/Engine/Core/PluginHost.hpp	37
ComponentType	Core	Source/Engine/Core/PluginHost.hpp	28
DateTime	Core	Source/Engine/Core/Calendar.hpp	46
DecimalStore	Core	Source/Engine/Core/DecimalStore.hpp	17
DecimalT	Core	Source/Engine/Core/	34

		Decimal.hpp	
DurationStyle	Core	Source/Engine/Core/Time.hpp	578
EddsaKeyPair	Core	Source/Engine/Core/CryptoAsymmetric.hpp	31
Error	Core	Source/Engine/Core/Error.hpp	23
ErrorCode	Core	Source/Engine/Core/Error.hpp	11
FieldDesc	Core	Source/Engine/Core/Mirror.hpp	40
FieldKind	Core	Source/Engine/Core/Mirror.hpp	17
FileFilter	Core	Source/Engine/Core/WindowManager.hpp	203
FormatSpec	Core	Source/Engine/Core/BigInt.hpp	36
FrameArena	Core	Source/Engine/Core/FrameArena.hpp	12
FrameTimer	Core	Source/Engine/Core/Time.hpp	508
GamepadState	Core	Source/Engine/Core/WindowManager.hpp	27
GridDesc	Core	Source/Engine/Core/Grid.hpp	91
Handle	Core	Source/Engine/Core/BigNumberStore.hpp	20
Handle	Core	Source/Engine/Core/Types.hpp	275
Handle	Core	Source/Engine/Core/WindowHost.hpp	20
Host	Core	Source/Engine/Core/Plugin.hpp	19
InputState	Core	Source/Engine/Core/WindowManager.hpp	126
JobSystem	Core	Source/Engine/Core/JobSystem.hpp	25
Key	Core	Source/Engine/	58

		Core/ WindowManager.hpp	
KeyLess	Core	Source/Engine/Core/ FlatMap.hpp	197
KeyUpperLess	Core	Source/Engine/Core/ FlatMap.hpp	209
LoadedPlugin	Core	Source/Engine/Core/ PluginHost.hpp	74
LogLevel	Core	Source/Engine/Core/ Log.hpp	13
LogRetention	Core	Source/Engine/Core/ Log.hpp	31
Mat4	Core	Source/Engine/ Core/Mat4.hpp	15
NoiseGenerator	Core	Source/Engine/Core/ NoiseGenerator.hpp	10
Notation	Core	Source/Engine/Core/ BigInt.hpp	30
NoticeEntry	Core	Source/Engine/Core/ ThirdPartyNotices.hpp	12
OptimizationAdvice	Core	Source/Engine/Core/ BuildProfile.hpp	154
OptimizationLevel	Core	Source/Engine/Core/ BuildProfile.hpp	10
PauseController	Core	Source/Engine/Core/ Pause.hpp	27
PauseSettings	Core	Source/Engine/Core/ Pause.hpp	19
Pcg32	Core	Source/Engine/Core/ Pcg32.hpp	20
PipelineTag	Core	Source/Engine/Core/ Types.hpp	303
PluginHost	Core	Source/Engine/Core/ PluginHost.hpp	86
PluginSystem	Core	Source/Engine/Core/ PluginHost.hpp	64
Profiler	Core	Source/Engine/Core/ Profiler.hpp	45
ProfileRow	Core	Source/Engine/Core/ ProfilerPanel.hpp	26
ProfilerPanel	Core	Source/Engine/Core/	39

		ProfilerPanel.hpp	
ProfileSample	Core	Source/Engine/Core/ Profiler.hpp	30
ProfileSortMode	Core	Source/Engine/Core/ ProfilerPanel.hpp	16
ProfileStats	Core	Source/Engine/Core/ Profiler.hpp	18
ProjectionMode	Core	Source/Engine/Core/ Grid.hpp	17
Rect	Core	Source/Engine/Core/ Types.hpp	173
ReducedAngle	Core	Source/Engine/Core/ SimMath.hpp	29
Reflection	Core	Source/Engine/ Core/Mirror.hpp	69
Reflection	Core	Source/Engine/ Core/Mirror.hpp	298
Reflection	Core	Source/Engine/ Core/Mirror.hpp	314
ReplayFrame	Core	Source/Engine/Core/ Replay.hpp	245
ReplayKeyframe	Core	Source/Engine/Core/ Replay.hpp	16
ReplaySession	Core	Source/Engine/Core/ Replay.hpp	25
RngState	Core	Source/Engine/Core/ Pcg32.hpp	11
RoundingMode	Core	Source/Engine/Core/ Decimal.hpp	22
SamplerTag	Core	Source/Engine/Core/ Types.hpp	304
Scope	Core	Source/Engine/Core/ Profiler.hpp	151
SecureRng	Core	Source/Engine/Core/ SecureRng.hpp	11
Sha256Context	Core	Source/Engine/Core/ Sha256.hpp	17
Slot	Core	Source/Engine/Core/ BigNumberStore.hpp	252
Slot	Core	Source/Engine/Core/ DecimalStore.hpp	199

SplashClock	Core	Source/Engine/Core/Splash.hpp	10
Status	Core	Source/Engine/Core/BigInt.hpp	18
Stopwatch	Core	Source/Engine/Core/Time.hpp	215
Task	Core	Source/Engine/Core/JobSystem.hpp	108
TextureTag	Core	Source/Engine/Core/Types.hpp	302
TextWriter	Core	Source/Engine/Core/Calendar.hpp	204
ThreadCounter	Core	Source/Engine/Core/JobSystem.hpp	121
ThreadLoad	Core	Source/Engine/Core/JobSystem.hpp	31
ThreadRole	Core	Source/Engine/Core/ThreadAffinity.hpp	12
TimerFired	Core	Source/Engine/Core/Time.hpp	261
TimerHandle	Core	Source/Engine/Core/Time.hpp	246
TimerPool	Core	Source/Engine/Core/Time.hpp	271
TouchPoint	Core	Source/Engine/Core/WindowManager.hpp	38
TypeNameOverride	Core	Source/Engine/Core/Definitions.hpp	42
TypeNameOverride	Core	Source/Engine/Core/Definitions.hpp	285
TypeObject	Core	Source/Engine/Core/Definitions.hpp	187
TypeRecord	Core	Source/Engine/Core/Definitions.hpp	87
TypeTable	Core	Source/Engine/Core/Definitions.hpp	95
TypeTraits	Core	Source/Engine/Core/Definitions.hpp	172
TypeTraits	Core	Source/Engine/Core/Definitions.hpp	269
Unregistered	Core	Source/Engine/	62

		Core/Mirror.hpp	
Vector2D	Core	Source/Engine/Core/Types.hpp	13
Vector3D	Core	Source/Engine/Core/Types.hpp	90
Vector4D	Core	Source/Engine/Core/Types.hpp	139
WindowDesc	Core	Source/Engine/Core/WindowManager.hpp	224
WindowExtent	Core	Source/Engine/Core/WindowManager.hpp	49
WindowHost	Core	Source/Engine/Core/WindowHost.hpp	17
WindowManager	Core	Source/Engine/Core/WindowManager.hpp	235
WindowMode	Core	Source/Engine/Core/WindowManager.hpp	215
WindowRecord	Core	Source/Engine/Core/WindowHost.hpp	30
X25519KeyPair	Core	Source/Engine/Core/CryptoAsymmetric.hpp	20

Engine/Documents (6)

Symbol	Kind	Header	Line
DocumentError	Documents	Source/Engine/Documents/Document.hpp	25
DocumentHandle	Documents	Source/Engine/Documents/Document.hpp	37
DocumentInfo	Documents	Source/Engine/Documents/Document.hpp	48
DocumentSystem	Documents	Source/Engine/Documents/Document.hpp	59
Slot	Documents	Source/Engine/Documents/	105

		Document.hpp	
VirtualFileSystem	Documents	Source/Engine/ Documents/ Document.hpp	14

Engine/ECS (289)

Symbol	Kind	Header	Line
AccessibilityKernel	ECS	Source/Engine/ECS/ SystemContext.hpp	16
AccessibilityServices	ECS	Source/Engine/ECS/ SystemContext.hpp	100
AccessibilitySystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	123
AccessibilitySystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	351
AccessValidation	ECS	Source/Engine/ECS/ FCS/ AccessValidation.hpp	17
AccessValidation	ECS	Source/Engine/ECS/ FCS/ AccessValidation.hpp	84
AICommandComponent	ECS	Source/Engine/ECS/ StateTree.hpp	175
AICommandKind	ECS	Source/Engine/ECS/ StateTree.hpp	163
AnimationClipBinding	ECS	Source/Engine/ECS/ AnimationClipPose.hpp	22
AnimationPose	ECS	Source/Engine/ECS/ AnimationClipPose.hpp	41
AnimationTimeline	ECS	Source/Engine/ECS/ AnimationTimeline.hpp	90
AnimBoneKey	ECS	Source/Engine/ECS/ AnimGraph.hpp	75
AnimClip	ECS	Source/Engine/ECS/ AnimGraph.hpp	87
AnimConditionFlags	ECS	Source/Engine/ECS/ AnimGraph.hpp	29
AnimGraphDefinitionComponent	ECS	Source/Engine/ECS/ AnimGraph.hpp	117

AnimGraphSystem	ECS	Source/Engine/ECS/AnimGraph.hpp	200
AnimRestBone	ECS	Source/Engine/ECS/AnimGraph.hpp	105
AnimStateComponent	ECS	Source/Engine/ECS/AnimGraph.hpp	138
AnimTransition	ECS	Source/Engine/ECS/AnimGraph.hpp	49
Archetype	ECS	Source/Engine/ECS/FCS/World.hpp	294
AssetDatabase	ECS	Source/Engine/ECS/Scene.hpp	21
AssetDatabase	ECS	Source/Engine/ECS/SystemContext.hpp	28
AssetServices	ECS	Source/Engine/ECS/SystemContext.hpp	71
AudioServices	ECS	Source/Engine/ECS/SystemContext.hpp	77
AudioSystem	ECS	Source/Engine/ECS/SystemContext.hpp	11
AudioSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	121
AudioSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	306
BigIntComponent	ECS	Source/Engine/ECS/Components.hpp	168
BoneAttachment2D	ECS	Source/Engine/ECS/BoneAttachment.hpp	19
BoneAttachmentSystem	ECS	Source/Engine/ECS/BoneAttachment.hpp	35
CachedTarget	ECS	Source/Engine/ECS/LightSource.hpp	234
Camera2D	ECS	Source/Engine/ECS/CoreComponents.hpp	90
CameraStackSystem	ECS	Source/Engine/ECS/SystemContext.hpp	21
CarrierMemberComponent	ECS	Source/Engine/ECS/Components.hpp	139
ChainIdTag	ECS	Source/Engine/ECS/StableIds.hpp	12
ChainSaveV3	ECS	Source/Engine/ECS/	28

		SnapshotWorld.hpp	
ChangeMaskBit	ECS	Source/Engine/ECS/ ChangeMask.hpp	20
CharacterDesc	ECS	Source/Engine/ECS/ Descriptors.hpp	51
Chunk	ECS	Source/Engine/ECS/ FCS/World.hpp	287
ChunkCopyRange	ECS	Source/Engine/ECS/ RenderEcs.hpp	90
ChunkHeader	ECS	Source/Engine/ECS/ RenderEcs.hpp	45
ChunkStorage	ECS	Source/Engine/ECS/ FCS/World.hpp	159
ClothIdTag	ECS	Source/Engine/ECS/ StableIds.hpp	13
ClothSaveV3	ECS	Source/Engine/ECS/ SnapshotWorld.hpp	59
ClothSystem	ECS	Source/Engine/ECS/ SystemContext.hpp	18
ColliderSnapshot	ECS	Source/Engine/ECS/ Snapshot.hpp	183
CollisionMode	ECS	Source/Engine/ECS/ Presentation.hpp	21
CollisionPolicyComponent	ECS	Source/Engine/ECS/ Presentation.hpp	101
Command	ECS	Source/Engine/ECS/ CommandBuffer.hpp	30
Command	ECS	Source/Engine/ECS/ FCS/ CommandBuffer.hpp	84
CommandBuffer	ECS	Source/Engine/ECS/ CommandBuffer.hpp	38
CommandBuffer	ECS	Source/Engine/ECS/ FCS/ CommandBuffer.hpp	20
CommandBuffer	ECS	Source/Engine/ECS/ SystemContext.hpp	49
CommandCutChain	ECS	Source/Engine/ECS/ CommandLog.hpp	42
CommandKind	ECS	Source/Engine/ECS/ CommandBuffer.hpp	20
CommandKind	ECS	Source/Engine/ECS/	75

		FCS/ CommandBuffer.hpp	
CommandLog	ECS	Source/Engine/ECS/ CommandLog.hpp	92
CommandLogHeader	ECS	Source/Engine/ECS/ CommandLog.hpp	66
CommandRecord	ECS	Source/Engine/ECS/ CommandLog.hpp	78
CommandType	ECS	Source/Engine/ECS/ CommandLog.hpp	22
CommandWorldCallbacks	ECS	Source/Engine/ECS/ CommandLog.hpp	52
ComponentChangeMask	ECS	Source/Engine/ECS/ ChangeMask.hpp	53
ComponentMeta	ECS	Source/Engine/ECS/ FCS/World.hpp	106
ComponentOps	ECS	Source/Engine/ECS/ ComponentRegistry.hpp	43
ComponentOpsFor	ECS	Source/Engine/ECS/ ComponentRegistry.hpp	68
ComponentRegistry	ECS	Source/Engine/ECS/ ComponentRegistry.hpp	129
ComponentStorage	ECS	Source/Engine/ECS/ FCS/Registry.hpp	775
CompressedClip	ECS	Source/Engine/ECS/ AnimationCompression.hpp	47
CompressedKey	ECS	Source/Engine/ECS/ AnimationCompression.hpp	19
CompressedTrack	ECS	Source/Engine/ECS/ AnimationCompression.hpp	31
CompressionReport	ECS	Source/Engine/ECS/ AnimationCompression.hpp	63
ConstCoreSystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	116
ConstCoreSystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	192

ConstRenderSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	119
ConstRenderSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	261
ConstView	ECS	Source/Engine/ECS/FCS/Registry.hpp	366
Context	ECS	Source/Engine/ECS/FCS/Registry.hpp	607
CoreServices	ECS	Source/Engine/ECS/SystemContext.hpp	55
CoreSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	115
CoreSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	168
Cursor	ECS	Source/Engine/ECS/CommandLog.hpp	177
Cursor	ECS	Source/Engine/ECS/FCS/Registry.hpp	114
Cursor	ECS	Source/Engine/ECS/FCS/Registry.hpp	374
DecimalComponent	ECS	Source/Engine/ECS/Components.hpp	179
DirectionalAnimationPlayer	ECS	Source/Engine/ECS/DirectionalAnimationPlayer.hpp	29
DirectionalBlendAnchor	ECS	Source/Engine/ECS/DirectionalBlend.hpp	52
DirectionalBlendAnchorEntry	ECS	Source/Engine/ECS/DirectionalBlend.hpp	84
DirectionalBlendSample	ECS	Source/Engine/ECS/DirectionalBlend.hpp	133
DirectionalBlendSpaceDocument	ECS	Source/Engine/ECS/DirectionalBlend.hpp	65
DirectionalBlendTable	ECS	Source/Engine/ECS/DirectionalBlend.hpp	106
DirectionalBlendTopology	ECS	Source/Engine/ECS/DirectionalBlend.hpp	42
DirectionalBlendTriangle	ECS	Source/Engine/ECS/DirectionalBlend.hpp	94
DirectionalBlendWeight	ECS	Source/Engine/ECS/DirectionalBlend.hpp	125
DrawOrder	ECS	Source/Engine/ECS/	79

		CoreComponents.hpp	
EachIterator	ECS	Source/Engine/ECS/FCS/Registry.hpp	224
EachIterator	ECS	Source/Engine/ECS/FCS/Registry.hpp	484
EachRange	ECS	Source/Engine/ECS/FCS/Registry.hpp	261
EachRange	ECS	Source/Engine/ECS/FCS/Registry.hpp	521
EditorSelected	ECS	Source/Engine/ECS/CoreComponents.hpp	104
EmitterIdTag	ECS	Source/Engine/ECS/StableIds.hpp	14
EmitterSaveV3	ECS	Source/Engine/ECS/SnapshotWorld.hpp	79
Entity	ECS	Source/Engine/ECS/FCS/World.hpp	48
EntityRecord	ECS	Source/Engine/ECS/FCS/World.hpp	712
EntityRemap	ECS	Source/Engine/ECS/Scene.hpp	59
EntityStorage	ECS	Source/Engine/ECS/FCS/Registry.hpp	691
Entry	ECS	Source/Engine/ECS/FCS/Registry.hpp	668
Entry	ECS	Source/Engine/ECS/Schedule.hpp	53
Facing2D	ECS	Source/Engine/ECS/DirectionalBlend.hpp	16
FeskIHeader	ECS	Source/Engine/ECS/SkeletonAsset.hpp	37
FeskINameRef	ECS	Source/Engine/ECS/SkeletonAsset.hpp	87
FeskISprite	ECS	Source/Engine/ECS/SkeletonAsset.hpp	97
FluidSaveV3	ECS	Source/Engine/ECS/SnapshotWorld.hpp	71
FluidWaveSystem	ECS	Source/Engine/ECS/SystemContext.hpp	19
FontAtlasSet	ECS	Source/Engine/ECS/SystemContext.hpp	36

FrameArena	ECS	Source/Engine/ECS/SystemContext.hpp	12
GameplayAttributeRegistry	ECS	Source/Engine/ECS/SystemContext.hpp	23
GameplayEffectSystem	ECS	Source/Engine/ECS/SystemContext.hpp	24
GameplayServices	ECS	Source/Engine/ECS/SystemContext.hpp	107
GameplaySkillSystem	ECS	Source/Engine/ECS/SystemContext.hpp	25
GameplaySystemView	ECS	Source/Engine/ECS/SystemContext.hpp	120
GameplaySystemView	ECS	Source/Engine/ECS/SystemContext.hpp	284
GameplayTagRegistry	ECS	Source/Engine/ECS/SystemContext.hpp	22
GpuEntity	ECS	Source/Engine/ECS/RenderEcs.hpp	31
GpuSpriteInstance	ECS	Source/Engine/ECS/RenderEcs.hpp	245
GraphSystem	ECS	Source/Engine/ECS/SystemGraph.hpp	91
Hidden	ECS	Source/Engine/ECS/CoreComponents.hpp	109
IkChain	ECS	Source/Engine/ECS/SkeletonIk.hpp	43
IkComponent	ECS	Source/Engine/ECS/SkeletonIk.hpp	64
IkGoalComponent	ECS	Source/Engine/ECS/SkeletonIk.hpp	77
IkSolver	ECS	Source/Engine/ECS/SkeletonIk.hpp	19
InputUserComponent	ECS	Source/Engine/ECS/Components.hpp	36
Interpolation	ECS	Source/Engine/ECS/AnimationTimeline.hpp	29
IterationGuard	ECS	Source/Engine/ECS/FCS/World.hpp	264
Iterator	ECS	Source/Engine/ECS/FCS/Registry.hpp	190
Iterator	ECS	Source/Engine/ECS/	450

		FCS/Registry.hpp	
Iterator	ECS	Source/Engine/ECS/FCS/Registry.hpp	698
ITextShaper	ECS	Source/Engine/ECS/SystemContext.hpp	42
Lifetime	ECS	Source/Engine/ECS/CoreComponents.hpp	98
Light2DComponent	ECS	Source/Engine/ECS/Lighting.hpp	37
Light2DOccluderComponent	ECS	Source/Engine/ECS/Lighting.hpp	109
Light2DResultComponent	ECS	Source/Engine/ECS/Lighting.hpp	69
Light2DTargetComponent	ECS	Source/Engine/ECS/Lighting.hpp	53
LightGrid2D	ECS	Source/Engine/ECS/LightGrid.hpp	26
LightGridInput	ECS	Source/Engine/ECS/LightGrid.hpp	16
LightingSettings	ECS	Source/Engine/ECS/Lighting.hpp	120
LightingSystem	ECS	Source/Engine/ECS/Lighting.hpp	147
LightSourceComponent	ECS	Source/Engine/ECS/LightSource.hpp	49
LightSourceFlags	ECS	Source/Engine/ECS/LightSource.hpp	33
LightSourceSettings	ECS	Source/Engine/ECS/LightSource.hpp	150
LightSourceSystem	ECS	Source/Engine/ECS/LightSource.hpp	169
LightSourceType	ECS	Source/Engine/ECS/LightSource.hpp	23
LocalizationTable	ECS	Source/Engine/ECS/SystemContext.hpp	17
LocalizedTextComponent	ECS	Source/Engine/ECS/Components.hpp	226
LocalTransform2D	ECS	Source/Engine/ECS/CoreComponents.hpp	27
Mat3	ECS	Source/Engine/ECS/Skeleton.hpp	32

NameComponent	ECS	Source/Engine/ECS/ CoreComponents.hpp	116
NarrativeLabelComponent	ECS	Source/Engine/ECS/ Components.hpp	157
NativeCleanupHook	ECS	Source/Engine/ECS/ RegistryOps.hpp	12
PackedTransform2D	ECS	Source/Engine/ECS/ RenderEcs.hpp	229
Parent2D	ECS	Source/Engine/ECS/ CoreComponents.hpp	35
Phase	ECS	Source/Engine/ECS/ Schedule.hpp	17
PhysicsBodyDesc	ECS	Source/Engine/ECS/ Descriptors.hpp	22
PhysicsJointSystem	ECS	Source/Engine/ECS/ SystemContext.hpp	20
PhysicsServices	ECS	Source/Engine/ECS/ SystemContext.hpp	82
PhysicsShapeKind	ECS	Source/Engine/ECS/ Descriptors.hpp	15
PhysicsSystem	ECS	Source/Engine/ECS/ SystemContext.hpp	13
PhysicsSystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	117
PhysicsSystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	216
PrefabAsset	ECS	Source/Engine/ECS/ Descriptors.hpp	88
PrefabEntityDesc	ECS	Source/Engine/ECS/ Descriptors.hpp	76
PrefabEntityFlags	ECS	Source/Engine/ECS/ Descriptors.hpp	98
PrefabEntityOverride	ECS	Source/Engine/ECS/ PrefabSystem.hpp	18
PrefabOverrides	ECS	Source/Engine/ECS/ PrefabSystem.hpp	31
PrefabSystem	ECS	Source/Engine/ECS/ PrefabSystem.hpp	36
QuantizedPose	ECS	Source/Engine/ECS/ SkeletonIk.hpp	165
Registry	ECS	Source/Engine/ECS/	804

		FCS/Registry.hpp	
RenderChunk	ECS	Source/Engine/ECS/ RenderEcs.hpp	62
RenderChunkArena	ECS	Source/Engine/ECS/ RenderEcs.hpp	304
RenderEcs	ECS	Source/Engine/ECS/ RenderEcs.hpp	460
RenderMaskComponent	ECS	Source/Engine/ECS/ Presentation.hpp	139
RenderPassMask	ECS	Source/Engine/ECS/ Presentation.hpp	126
RenderServices	ECS	Source/Engine/ECS/ SystemContext.hpp	64
RenderSystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	118
RenderSystemView	ECS	Source/Engine/ECS/ SystemContext.hpp	239
ReplayPlayback	ECS	Source/Engine/ECS/ ReplayRecorder.hpp	103
ReplayRecorder	ECS	Source/Engine/ECS/ ReplayRecorder.hpp	32
RetargetMap	ECS	Source/Engine/ECS/ SkeletonIk.hpp	131
RetargetPair	ECS	Source/Engine/ECS/ SkeletonIk.hpp	118
RHI	ECS	Source/Engine/ECS/ SystemContext.hpp	32
RigidBodySnapshot	ECS	Source/Engine/ECS/ Snapshot.hpp	169
SceneComponent	ECS	Source/Engine/ECS/ Scene.hpp	38
SceneData	ECS	Source/Engine/ECS/ Scene.hpp	52
SceneEntity	ECS	Source/Engine/ECS/ Scene.hpp	45
SceneLoadOptions	ECS	Source/Engine/ECS/ Scene.hpp	76
SceneMeta	ECS	Source/Engine/ECS/ Scene.hpp	31
Schedule	ECS	Source/Engine/ECS/ Schedule.hpp	48

ScopedAccess	ECS	Source/Engine/ECS/FCS/AccessValidation.hpp	104
SdfBrickMap	ECS	Source/Engine/ECS/Lighting.hpp	17
SdfGiVolume	ECS	Source/Engine/ECS/Lighting.hpp	18
SdfLightingFrame	ECS	Source/Engine/ECS/Lighting.hpp	22
SdfSunComponent	ECS	Source/Engine/ECS/Lighting.hpp	84
SkeletonAssetRef	ECS	Source/Engine/ECS/SkeletonAssetRef.hpp	20
SkeletonAssetSource	ECS	Source/Engine/ECS/SkeletonAsset.hpp	189
SkeletonAssetView	ECS	Source/Engine/ECS/SkeletonAsset.hpp	121
SkeletonBoneName	ECS	Source/Engine/ECS/SkeletonAsset.hpp	168
SkeletonComponent	ECS	Source/Engine/ECS/Skeleton.hpp	122
SkeletonNameTable	ECS	Source/Engine/ECS/SkeletonAsset.hpp	175
SkeletonSystem	ECS	Source/Engine/ECS/Skeleton.hpp	205
SkinComponent	ECS	Source/Engine/ECS/Skeleton.hpp	143
SkinnedSprite	ECS	Source/Engine/ECS/Skeleton.hpp	154
SleepingStaticComponent	ECS	Source/Engine/ECS/Presentation.hpp	202
SnapshotBaseline	ECS	Source/Engine/ECS/Snapshot.hpp	316
SnapshotComponent	ECS	Source/Engine/ECS/Snapshot.hpp	78
SnapshotComponentOp	ECS	Source/Engine/ECS/Snapshot.hpp	43
SnapshotExtras	ECS	Source/Engine/ECS/Snapshot.hpp	267
SnapshotFlags	ECS	Source/Engine/ECS/Snapshot.hpp	31
SnapshotHeader	ECS	Source/Engine/ECS/	49

		Snapshot.hpp	
SnapshotRecordFlags	ECS	Source/Engine/ECS/Snapshot.hpp	37
SnapshotRestoreExtras	ECS	Source/Engine/ECS/Snapshot.hpp	272
SnapshotSchema	ECS	Source/Engine/ECS/Snapshot.hpp	93
SnapshotSchema	ECS	Source/Engine/ECS/Snapshot.hpp	208
SnapshotView	ECS	Source/Engine/ECS/Snapshot.hpp	257
Sprite2D	ECS	Source/Engine/ECS/CoreComponents.hpp	47
SpriteAnimationComponent	ECS	Source/Engine/ECS/SpriteAnimation.hpp	21
SpriteAnimationEventQueue	ECS	Source/Engine/ECS/SpriteAnimation.hpp	74
SpriteAnimationSystem	ECS	Source/Engine/ECS/SpriteAnimation.hpp	118
SpriteFrameEvent	ECS	Source/Engine/ECS/SpriteAnimation.hpp	66
SpriteFrameRect	ECS	Source/Engine/ECS/SpriteAnimation.hpp	55
SpriteOverlay2D	ECS	Source/Engine/ECS/CoreComponents.hpp	64
SpriteSheetComponent	ECS	Source/Engine/ECS/SpriteAnimation.hpp	40
StableIdAllocator	ECS	Source/Engine/ECS/StableIds.hpp	25
StateTreeComponent	ECS	Source/Engine/ECS/StateTree.hpp	195
StateTreeDefinition	ECS	Source/Engine/ECS/StateTree.hpp	152
StateTreeEvaluator	ECS	Source/Engine/ECS/StateTree.hpp	55
StateTreeEvaluatorKind	ECS	Source/Engine/ECS/StateTree.hpp	28
StateTreeNode	ECS	Source/Engine/ECS/StateTree.hpp	115
StateTreeNodeKind	ECS	Source/Engine/ECS/StateTree.hpp	105

StateTreeSystem	ECS	Source/Engine/ECS/StateTree.hpp	239
StateTreeTask	ECS	Source/Engine/ECS/StateTree.hpp	87
StateTreeTaskKind	ECS	Source/Engine/ECS/StateTree.hpp	73
StateTreeTransition	ECS	Source/Engine/ECS/StateTree.hpp	134
SystemAccess	ECS	Source/Engine/ECS/SystemGraph.hpp	65
SystemContext	ECS	Source/Engine/ECS/SystemContext.hpp	129
SystemGraph	ECS	Source/Engine/ECS/Schedule.hpp	15
SystemGraph	ECS	Source/Engine/ECS/SystemGraph.hpp	116
TargetIndexEntry	ECS	Source/Engine/ECS/LightSource.hpp	244
TextComponent	ECS	Source/Engine/ECS/Components.hpp	198
TextConditionComponent	ECS	Source/Engine/ECS/Components.hpp	282
TextEffectComponent	ECS	Source/Engine/ECS/Components.hpp	252
TextFormatComponent	ECS	Source/Engine/ECS/Components.hpp	264
TextService	ECS	Source/Engine/ECS/SystemContext.hpp	40
TextServices	ECS	Source/Engine/ECS/SystemContext.hpp	90
TextShapeCache	ECS	Source/Engine/ECS/SystemContext.hpp	41
TextShapeComponent	ECS	Source/Engine/ECS/Components.hpp	241
TilemapAssetRefComponent	ECS	Source/Engine/ECS/Tilemap.hpp	130
TilemapComponent	ECS	Source/Engine/ECS/Tilemap.hpp	63
TilemapRenderSystem	ECS	Source/Engine/ECS/Tilemap.hpp	150
TileQuad	ECS	Source/Engine/ECS/	142

		Tilemap.hpp	
TimelineEvent	ECS	Source/Engine/ECS/AnimationTimeline.hpp	67
TimelineEventKind	ECS	Source/Engine/ECS/AnimationTimeline.hpp	36
TimelineKey	ECS	Source/Engine/ECS/AnimationTimeline.hpp	43
TimelineTrack	ECS	Source/Engine/ECS/AnimationTimeline.hpp	56
TimelineTriggerSink	ECS	Source/Engine/ECS/AnimationTimeline.hpp	83
TimeScaleComponent	ECS	Source/Engine/ECS/Presentation.hpp	168
TrackKind	ECS	Source/Engine/ECS/AnimationTimeline.hpp	20
Transform2D	ECS	Source/Engine/ECS/CoreComponents.hpp	16
TransportGroupComponent	ECS	Source/Engine/ECS/Components.hpp	123
TransportMotion	ECS	Source/Engine/ECS/Components.hpp	109
TransportPath	ECS	Source/Engine/ECS/Components.hpp	100
TransportSurface	ECS	Source/Engine/ECS/Components.hpp	90
UiSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	122
UiSystemView	ECS	Source/Engine/ECS/SystemContext.hpp	326
UvCacheEntry	ECS	Source/Engine/ECS/Tilemap.hpp	176
Vehicle2D	ECS	Source/Engine/ECS/Components.hpp	54
VehicleFlags	ECS	Source/Engine/ECS/Components.hpp	46
Velocity	ECS	Source/Engine/ECS/CoreComponents.hpp	41

View	ECS	Source/Engine/ECS/ CommandLog.hpp	187
View	ECS	Source/Engine/ECS/ FCS/Registry.hpp	106
ViewIterationGuard	ECS	Source/Engine/ECS/ FCS/Registry.hpp	41
VirtualFileSystem	ECS	Source/Engine/ECS/ DirectionalAnimationPl ayer.hpp	16
VirtualFileSystem	ECS	Source/Engine/ECS/ Scene.hpp	18
VirtualFileSystem	ECS	Source/Engine/ECS/ SkeletonAsset.hpp	18
VirtualFileSystem	ECS	Source/Engine/ECS/ SystemContext.hpp	15
VirtualFileSystem	ECS	Source/Engine/ECS/ Tilemap.hpp	20
Wheel2D	ECS	Source/Engine/ECS/ Components.hpp	70
WindowManager	ECS	Source/Engine/ECS/ SystemContext.hpp	14
World	ECS	Source/Engine/ECS/ FCS/World.hpp	284
WorldMemoryStats	ECS	Source/Engine/ECS/ FCS/World.hpp	241
WorldSnapshotV3	ECS	Source/Engine/ECS/ SnapshotWorld.hpp	86

Engine/Effects (16)

Symbol	Kind	Header	Line
Binding	Effects	Source/Engine/ Effects/ ParticleScene.hpp	93
EmitterState	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	174
ParticleEffectCompo nent	Effects	Source/Engine/ Effects/ ParticleScene.hpp	24
ParticleEffectCompo nentFlags	Effects	Source/Engine/ Effects/ ParticleScene.hpp	17

ParticleEffectPreset	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	120
ParticleEffectSettings	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	70
ParticleEffectSystem	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	140
ParticleEmissionShape	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	42
ParticlePresetDesc	Effects	Source/Engine/ Effects/ ParticlePreset.hpp	26
ParticlePresetFlags	Effects	Source/Engine/ Effects/ ParticlePreset.hpp	20
ParticlePresetRegistry	Effects	Source/Engine/ Effects/ ParticlePreset.hpp	38
ParticleSceneSystem	Effects	Source/Engine/ Effects/ ParticleScene.hpp	74
ParticleShape	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	23
ParticleShapeAtlas	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	58
ParticleShapeFrame	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	51
ParticleState	Effects	Source/Engine/ Effects/ ParticleEffects.hpp	187

Engine/Gameplay (32)

Symbol	Kind	Header	Line
ActiveGameplayEffect	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	66
DefIndexEntry	Gameplay	Source/Engine/ Gameplay/	163

		GameplayEffects.hpp	
Entry	Gameplay	Source/Engine/ Gameplay/ GameplayAttributes.hpp	50
Entry	Gameplay	Source/Engine/ Gameplay/ GameplayTags.hpp	152
GameplayAttribute	Gameplay	Source/Engine/ Gameplay/ GameplayAttributes.hpp	21
GameplayAttributeModifier	Gameplay	Source/Engine/ Gameplay/ GameplayAttributes.hpp	82
GameplayAttributeRegistry	Gameplay	Source/Engine/ Gameplay/ GameplayAttributes.hpp	35
GameplayAttributesComponent	Gameplay	Source/Engine/ Gameplay/ GameplayAttributes.hpp	66
GameplayAttributeValues	Gameplay	Source/Engine/ Gameplay/ GameplayAttributes.hpp	60
GameplayEffectDef	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	45
GameplayEffectEvent	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	95
GameplayEffectEventType	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	88
GameplayEffectFlags	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	32
GameplayEffectsComponent	Gameplay	Source/Engine/ Gameplay/ GameplayEffects.hpp	78
GameplayEffectStackPolicy	Gameplay	Source/Engine/ Gameplay/	22

		GameplayEffects.hpp	
GameplayEffectSystem	Gameplay	Source/Engine/Gameplay/GameplayEffects.hpp	107
GameplayModifierOp	Gameplay	Source/Engine/Gameplay/GameplayAttributes.hpp	75
GameplaySkillDef	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	55
GameplaySkillEvent	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	98
GameplaySkillEventType	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	33
GameplaySkillFailReason	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	41
GameplaySkillsComponent	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	89
GameplaySkillSlot	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	77
GameplaySkillState	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	27
GameplaySkillSystem	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	110
GameplaySkillTargetKind	Gameplay	Source/Engine/Gameplay/GameplaySkills.hpp	19
GameplayTag	Gameplay	Source/Engine/Gameplay/GameplayTags.hpp	22
GameplayTagContainer	Gameplay	Source/Engine/Gameplay/GameplayTags.hpp	170
GameplayTagDatabase	Gameplay	Source/Engine/Gameplay/GameplayTagDatabase.hpp	18

GameplayTagMask	Gameplay	Source/Engine/ Gameplay/ GameplayTags.hpp	38
GameplayTagRegistry	Gameplay	Source/Engine/ Gameplay/ GameplayTags.hpp	132
GameplayTagsComponent	Gameplay	Source/Engine/ Gameplay/ GameplayTags.hpp	211

Engine/Input (14)

Symbol	Kind	Header	Line
InputActionDesc	Input	Source/Engine/Input/ InputSystem.hpp	83
InputActionState	Input	Source/Engine/Input/ InputSystem.hpp	110
InputActionType	Input	Source/Engine/Input/ InputSystem.hpp	23
InputContextDesc	Input	Source/Engine/Input/ InputSystem.hpp	89
InputEvent	Input	Source/Engine/Input/ InputSystem.hpp	122
InputEventType	Input	Source/Engine/Input/ InputSystem.hpp	62
InputMappingDesc	Input	Source/Engine/Input/ InputSystem.hpp	96
InputModifier	Input	Source/Engine/Input/ InputSystem.hpp	76
InputModifierKind	Input	Source/Engine/Input/ InputSystem.hpp	41
InputSource	Input	Source/Engine/Input/ InputSystem.hpp	70
InputSourceKind	Input	Source/Engine/Input/ InputSystem.hpp	30
InputSystem	Input	Source/Engine/Input/ InputSystem.hpp	134
InputTriggerKind	Input	Source/Engine/Input/ InputSystem.hpp	52
UserSlot	Input	Source/Engine/Input/ InputSystem.hpp	180

Engine/Navigation (25)

Symbol	Kind	Header	Line
CrowdAgent	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	16
CrowdPool	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	46
CrowdRenderSettings	Navigation	Source/Engine/ Navigation/ CrowdRender.hpp	14
CrowdSettings	Navigation	Source/Engine/ Navigation/ CrowdPool.hpp	27
Entry	Navigation	Source/Engine/ Navigation/ FlowField.hpp	79
FlowField	Navigation	Source/Engine/ Navigation/ FlowField.hpp	33
FlowFieldDesc	Navigation	Source/Engine/ Navigation/ FlowField.hpp	22
NavAgentComponent	Navigation	Source/Engine/ Navigation/ Navigation.hpp	242
NavAgentPathSettings	Navigation	Source/Engine/ Navigation/ Navigation.hpp	206
NavAgentSettings	Navigation	Source/Engine/ Navigation/ Navigation.hpp	184
NavigationSystem	Navigation	Source/Engine/ Navigation/ Navigation.hpp	267
NavMap	Navigation	Source/Engine/ Navigation/ Navigation.hpp	27
NavMesh	Navigation	Source/Engine/ Navigation/ NavMesh.hpp	49
NavMesh	Navigation	Source/Engine/ Navigation/ Navigation.hpp	18

NavMeshPortal	Navigation	Source/Engine/ Navigation/ NavMesh.hpp	39
NavMeshRect	Navigation	Source/Engine/ Navigation/ NavMesh.hpp	19
NavPath	Navigation	Source/Engine/ Navigation/ Navigation.hpp	99
NavPathOptions	Navigation	Source/Engine/ Navigation/ Navigation.hpp	80
NavSmoothingMode	Navigation	Source/Engine/ Navigation/ Navigation.hpp	72
Page	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	105
PagedNavGrid	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	32
PagedNavGridDesc	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	23
PathFinder	Navigation	Source/Engine/ Navigation/ Navigation.hpp	127
PathStats	Navigation	Source/Engine/ Navigation/ Navigation.hpp	116
TilemapComponent	Navigation	Source/Engine/ Navigation/ PagedNavGrid.hpp	14

Engine/Network (91)

Symbol	Kind	Header	Line
CellCoordinate	Network	Source/Engine/ Network/ RelevanceGrid.hpp	168
ChecksumComparison	Network	Source/Engine/ Network/ NetRollback.hpp	457
Config	Network	Source/Engine/ Network/	316

		NetRollback.hpp	
DatagramSocket	Network	Source/Engine/ Network/ Transport.hpp	32
Endpoint	Network	Source/Engine/ Network/NetLink.hpp	92
Frame	Network	Source/Engine/ Network/NetLink.hpp	84
FrameInfo	Network	Source/Engine/ Network/ NetRollback.hpp	221
FrameMeta	Network	Source/Engine/ Network/ NetRollback.hpp	473
InputChannel	Network	Source/Engine/ Network/ NetProtocol.hpp	56
InputChannelStats	Network	Source/Engine/ Network/ NetProtocol.hpp	44
InputPacket	Network	Source/Engine/ Network/ NetProtocol.hpp	24
InputRingBuffer	Network	Source/Engine/ Network/ NetRollback.hpp	123
LobbyError	Network	Source/Engine/ Network/Lobby.hpp	37
LobbyMessageType	Network	Source/Engine/ Network/Lobby.hpp	25
LobbyRoomInfo	Network	Source/Engine/ Network/Lobby.hpp	73
LobbyServer	Network	Source/Engine/ Network/Lobby.hpp	93
LobbyServerConfig	Network	Source/Engine/ Network/Lobby.hpp	82
LocalEntry	Network	Source/Engine/ Network/ NetRollback.hpp	205
Member	Network	Source/Engine/ Network/Lobby.hpp	126
MemoryEndpoint	Network	Source/Engine/ Network/NetLink.hpp	46

MemoryEndpoint	Network	Source/Engine/ Network/NetLink.hpp	113
MemoryNetwork	Network	Source/Engine/ Network/NetLink.hpp	53
NetAddress	Network	Source/Engine/ Network/ Transport.hpp	19
NetDiagnosticsSnapshot	Network	Source/Engine/ Network/ Diagnostics.hpp	12
NetDisconnectReason	Network	Source/Engine/ Network/Session.hpp	65
NetEntityId	Network	Source/Engine/ Network/ NetRollback.hpp	39
NetEntityMap	Network	Source/Engine/ Network/ NetRollback.hpp	167
NetEntry	Network	Source/Engine/ Network/ NetRollback.hpp	200
NetIdentity	Network	Source/Engine/ Network/Identity.hpp	32
NetInputFrame	Network	Source/Engine/ Network/ NetRollback.hpp	70
NetInputStatus	Network	Source/Engine/ Network/ NetRollback.hpp	99
NetLink	Network	Source/Engine/ Network/NetLink.hpp	19
NetLobbyClient	Network	Source/Engine/ Network/Lobby.hpp	160
NetLobbyClientConfig	Network	Source/Engine/ Network/Lobby.hpp	151
NetPlayerId	Network	Source/Engine/ Network/ NetRollback.hpp	59
NetPlayerSlot	Network	Source/Engine/ Network/Session.hpp	134
NetRole	Network	Source/Engine/ Network/Session.hpp	32
NetRollbackSession	Network	Source/Engine/ Network/	240

		NetRollback.hpp	
NetSession	Network	Source/Engine/ Network/Session.hpp	189
NetSessionCallbacks	Network	Source/Engine/ Network/Session.hpp	178
NetSessionConfig	Network	Source/Engine/ Network/Lobby.hpp	15
NetSessionConfig	Network	Source/Engine/ Network/Session.hpp	149
NetSessionMode	Network	Source/Engine/ Network/Session.hpp	22
NetSessionStatus	Network	Source/Engine/ Network/Session.hpp	46
NetSlotState	Network	Source/Engine/ Network/Session.hpp	58
NetTicket	Network	Source/Engine/ Network/Ticket.hpp	35
NetTickInputs	Network	Source/Engine/ Network/ NetRollback.hpp	110
NetTickMode	Network	Source/Engine/ Network/Session.hpp	40
NetTopology	Network	Source/Engine/ Network/ NetTopology.hpp	11
NetTopologyPlan	Network	Source/Engine/ Network/ NetTopology.hpp	24
NetTransport	Network	Source/Engine/ Network/ Transport.hpp	130
Outstanding	Network	Source/Engine/ Network/ Transport.hpp	103
Peer	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	87
PendingEvent	Network	Source/Engine/ Network/Session.hpp	279
PendingState	Network	Source/Engine/ Network/ Replication.hpp	129

Phase	Network	Source/Engine/ Network/ SecureChannel.hpp	256
PhysicsSystem	Network	Source/Engine/ Network/ NetRollback.hpp	20
RateEntry	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	120
Reassembly	Network	Source/Engine/ Network/ Transport.hpp	115
RelevanceBand	Network	Source/Engine/ Network/ RelevanceGrid.hpp	32
RelevanceCell	Network	Source/Engine/ Network/ RelevanceGrid.hpp	71
RelevanceConfig	Network	Source/Engine/ Network/ NetRollback.hpp	34
RelevanceConfig	Network	Source/Engine/ Network/ RelevanceGrid.hpp	56
RelevanceEntity	Network	Source/Engine/ Network/ RelevanceGrid.hpp	81
RelevanceGrid	Network	Source/Engine/ Network/ NetRollback.hpp	36
RelevanceGrid	Network	Source/Engine/ Network/ RelevanceGrid.hpp	131
RelevanceGridStats	Network	Source/Engine/ Network/ RelevanceGrid.hpp	121
RelevancePlan	Network	Source/Engine/ Network/ NetRollback.hpp	35
RelevancePlan	Network	Source/Engine/ Network/ RelevanceGrid.hpp	93
ReplicationStatus	Network	Source/Engine/ Network/	28

		Replication.hpp	
Result	Network	Source/Engine/ Network/ NetRollback.hpp	243
RollbackReport	Network	Source/Engine/ Network/ NetRollback.hpp	229
Room	Network	Source/Engine/ Network/Lobby.hpp	132
SecureChannel	Network	Source/Engine/ Network/ SecureChannel.hpp	198
SecureChannelConfig	Network	Source/Engine/ Network/ SecureChannel.hpp	151
SecureCookieIssuer	Network	Source/Engine/ Network/ SecureChannel.hpp	131
SecureEndpoint	Network	Source/Engine/ Network/ SecureEndpoint.hpp	16
SecureSessionAuthConfig	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	24
SecureSessionTransport	Network	Source/Engine/ Network/ SecureSessionTransport.hpp	38
SecureStatus	Network	Source/Engine/ Network/ SecureChannel.hpp	65
Slot	Network	Source/Engine/ Network/ NetRollback.hpp	155
SnapshotReceiver	Network	Source/Engine/ Network/ Replication.hpp	177
SnapshotReceiverConfig	Network	Source/Engine/ Network/ Replication.hpp	166
SnapshotSender	Network	Source/Engine/ Network/ Replication.hpp	94
SnapshotSenderCon	Network	Source/Engine/	70

fig		Network/ Replication.hpp	
SnapshotSenderStats	Network	Source/Engine/ Network/ Replication.hpp	80
SortableEntity	Network	Source/Engine/ Network/ RelevanceGrid.hpp	195
TicketStatus	Network	Source/Engine/ Network/Ticket.hpp	54
TransportChannel	Network	Source/Engine/ Network/ Transport.hpp	63
TransportPeer	Network	Source/Engine/ Network/ Transport.hpp	85
TransportStats	Network	Source/Engine/ Network/ Transport.hpp	69

Engine/Physics (142)

Symbol	Kind	Header	Line
BodyColumns	Physics	Source/Engine/ Physics/ DodPhysicsBackend.hpp	59
BodyRecord	Physics	Source/Engine/ Physics/Physics.hpp	307
BodyType	Physics	Source/Engine/ Physics/Physics.hpp	53
ChainBreakEvent	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	314
ChainError	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	146
ChainRuntime	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	618
ClothComponent	Physics	Source/Engine/ Physics/Cloth.hpp	31
ClothDesc	Physics	Source/Engine/ Physics/Cloth.hpp	63

ClothFlags	Physics	Source/Engine/ Physics/Cloth.hpp	20
ClothHandle	Physics	Source/Engine/ Physics/Cloth.hpp	46
ClothInfo	Physics	Source/Engine/ Physics/Cloth.hpp	99
ClothRenderSystem	Physics	Source/Engine/ Physics/ ClothRenderSystem.hpp	15
ClothSystem	Physics	Source/Engine/ Physics/Cloth.hpp	110
Collider2D	Physics	Source/Engine/ Physics/Physics.hpp	90
CollisionFilterMatrix	Physics	Source/Engine/ Physics/ CollisionFilter.hpp	48
CollisionLayer	Physics	Source/Engine/ Physics/ CollisionFilter.hpp	15
ConnectionRuntime	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	590
ContactRecord	Physics	Source/Engine/ Physics/ DodPhysicsBackend.hpp	130
DodPhysicsBackend	Physics	Source/Engine/ Physics/ DodPhysicsBackend.hpp	16
DodPhysicsBackend	Physics	Source/Engine/ Physics/Physics.hpp	20
FluidDesc	Physics	Source/Engine/ Physics/Fluid.hpp	114
FluidInteractorComponent	Physics	Source/Engine/ Physics/Fluid.hpp	210
FluidWaveSystem	Physics	Source/Engine/ Physics/Fluid.hpp	221
GerstnerField	Physics	Source/Engine/ Physics/Fluid.hpp	49
GerstnerSample	Physics	Source/Engine/ Physics/Fluid.hpp	39
GerstnerWave	Physics	Source/Engine/	25

		Physics/Fluid.hpp	
GravityBodyComponent	Physics	Source/Engine/Physics/Gravity.hpp	79
GravityBodyMode	Physics	Source/Engine/Physics/Gravity.hpp	33
GravityBodyShape	Physics	Source/Engine/Physics/Gravity.hpp	41
GravityDirection	Physics	Source/Engine/Physics/Gravity.hpp	15
GravityFalloff	Physics	Source/Engine/Physics/Gravity.hpp	25
GravitySystem	Physics	Source/Engine/Physics/Gravity.hpp	124
GridEntry	Physics	Source/Engine/Physics/DodPhysicsBackend.hpp	121
GroupIndexEntry	Physics	Source/Engine/Physics/TransportGroups.hpp	127
JointBreakReason	Physics	Source/Engine/Physics/PhysicsJoints.hpp	72
JointChainComponent	Physics	Source/Engine/Physics/PhysicsJoints.hpp	378
JointChainDesc	Physics	Source/Engine/Physics/PhysicsJoints.hpp	187
JointChainFlags	Physics	Source/Engine/Physics/PhysicsJoints.hpp	91
JointChainHandle	Physics	Source/Engine/Physics/PhysicsJoints.hpp	250
JointChainInfo	Physics	Source/Engine/Physics/PhysicsJoints.hpp	429
JointChainKind	Physics	Source/Engine/Physics/PhysicsJoints.hpp	54
JointChainSection	Physics	Source/Engine/Physics/PhysicsJoints.hpp	119

JointConnectionFlags	Physics	Source/Engine/Physics/PhysicsJoints.hpp	80
JointConnectionHandle	Physics	Source/Engine/Physics/PhysicsJoints.hpp	268
JointConnectionInfo	Physics	Source/Engine/Physics/PhysicsJoints.hpp	287
JointConnectionKind	Physics	Source/Engine/Physics/PhysicsJoints.hpp	64
JointConnectionParams	Physics	Source/Engine/Physics/PhysicsJoints.hpp	559
JointLinkComponent	Physics	Source/Engine/Physics/PhysicsJoints.hpp	33
JointLinkFlags	Physics	Source/Engine/Physics/PhysicsJoints.hpp	22
JointTieDesc	Physics	Source/Engine/Physics/PhysicsJoints.hpp	328
JointTieSave	Physics	Source/Engine/Physics/PhysicsJoints.hpp	351
LevelGravityComponent	Physics	Source/Engine/Physics/Gravity.hpp	55
ParticleEmitterComponent	Physics	Source/Engine/Physics/Particles.hpp	34
ParticleEmitterFlags	Physics	Source/Engine/Physics/Particles.hpp	18
ParticleEmitterGpu	Physics	Source/Engine/Physics/Particles.hpp	69
ParticleSeedPush	Physics	Source/Engine/Physics/Particles.hpp	92
ParticleStateGpu	Physics	Source/Engine/Physics/Particles.hpp	100
PhysicsBackendKind	Physics	Source/Engine/Physics/Physics.hpp	32
PhysicsBodyRemapEntry	Physics	Source/Engine/Physics/SnapshotRemap.hpp	17

PhysicsBodyRemapTable	Physics	Source/Engine/Physics/SnapshotRemap.hpp	24
PhysicsCapabilities	Physics	Source/Engine/Physics/Physics.hpp	39
PhysicsContactEvent	Physics	Source/Engine/Physics/Physics.hpp	118
PhysicsDesc	Physics	Source/Engine/Physics/Physics.hpp	154
PhysicsEventType	Physics	Source/Engine/Physics/Physics.hpp	111
PhysicsFixedStepSystem	Physics	Source/Engine/Physics/Physics.hpp	340
PhysicsJointSystem	Physics	Source/Engine/Physics/PhysicsJoints.hpp	444
PhysicsRayHit	Physics	Source/Engine/Physics/Physics.hpp	129
PhysicsSystem	Physics	Source/Engine/Physics/Physics.hpp	168
PhysicsSystem	Physics	Source/Engine/Physics/TransportGroups.hpp	13
Projectile2D	Physics	Source/Engine/Physics/Physics.hpp	143
ProjectileHitEvent	Physics	Source/Engine/Physics/ProjectileSystem.hpp	12
ProjectileSystem	Physics	Source/Engine/Physics/ProjectileSystem.hpp	24
RigidBody2D	Physics	Source/Engine/Physics/Physics.hpp	73
RigidBodyFlags	Physics	Source/Engine/Physics/Physics.hpp	60
Runtime	Physics	Source/Engine/Physics/Cloth.hpp	156
SdfAtlasImage	Physics	Source/Engine/Physics/SdfAtlas.hpp	18
SdfBonePart	Physics	Source/Engine/Physics/SdfCharacter.hpp	24
SdfBrickGpu	Physics	Source/Engine/	48

		Physics/ SdfBrickMap.hpp	
SdfBrickHeader	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	32
SdfBrickHeader	Physics	Source/Engine/ Physics/ SdfCollision.hpp	107
SdfBrickMap	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	79
SdfBrickMap	Physics	Source/Engine/ Physics/ SdfCollision.hpp	106
SdfBrickMapDesc	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	63
SdfBrickState	Physics	Source/Engine/ Physics/ SdfBrickMap.hpp	24
SdfCarveOp	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	25
SdfCarveReplayContext	Physics	Source/Engine/ Physics/ SdfDestructionCommands.hpp	37
SdfCarveShape	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	17
SdfCharacterComponent	Physics	Source/Engine/ Physics/ SdfCharacter.hpp	41
SdfCharacterField	Physics	Source/Engine/ Physics/ SdfCharacter.hpp	78
SdfCharacterMeshData	Physics	Source/Engine/ Physics/ SdfCharacter.hpp	110
SdfCollisionDesc	Physics	Source/Engine/ Physics/ SdfCollision.hpp	49
SdfCollisionSystem	Physics	Source/Engine/ Physics/ SdfCollision.hpp	109

SdfComposeLayer	Physics	Source/Engine/ Physics/ SdfComposer.hpp	25
SdfComposeOp	Physics	Source/Engine/ Physics/ SdfComposer.hpp	16
SdfComposer	Physics	Source/Engine/ Physics/ SdfComposer.hpp	37
SdfComposerSession	Physics	Source/Engine/ Physics/ SdfComposerSession.hpp	15
SdfConnection	Physics	Source/Engine/ Physics/ SdfDamage.hpp	67
SdfContact	Physics	Source/Engine/ Physics/ SdfCollision.hpp	66
SdfContactManifold	Physics	Source/Engine/ Physics/ SdfCollision.hpp	97
SdfContactPoint	Physics	Source/Engine/ Physics/ SdfCollision.hpp	86
SdfDamageModel	Physics	Source/Engine/ Physics/ SdfDamage.hpp	189
SdfDebrisPiece	Physics	Source/Engine/ Physics/SdfDebris.hpp	20
SdfDebrisSettings	Physics	Source/Engine/ Physics/SdfDebris.hpp	31
SdfDestructible	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	56
SdfDestructible	Physics	Source/Engine/ Physics/SdfWorld.hpp	16
SdfDestructibleDesc	Physics	Source/Engine/ Physics/ SdfDestructible.hpp	36
SdfField	Physics	Source/Engine/ Physics/ SdfCollision.hpp	19
SdfGiAsset	Physics	Source/Engine/ Physics/	65

		SdfGiAsset.hpp	
SdfGiAssetHeader	Physics	Source/Engine/ Physics/ SdfGiAsset.hpp	45
SdfGiBaked	Physics	Source/Engine/ Physics/ SdfGiBake.hpp	45
SdfGiBakedProbe	Physics	Source/Engine/ Physics/ SdfGiBake.hpp	33
SdfGiBakeSettings	Physics	Source/Engine/ Physics/ SdfGiBake.hpp	17
SdfGiProbe	Physics	Source/Engine/ Physics/SdfGi.hpp	22
SdfGiVolume	Physics	Source/Engine/ Physics/SdfGi.hpp	41
SdfImpact	Physics	Source/Engine/ Physics/ SdfDamage.hpp	86
SdfImpactResult	Physics	Source/Engine/ Physics/ SdfDamage.hpp	101
SdfMaterialProfile	Physics	Source/Engine/ Physics/ SdfDamage.hpp	26
SdfPart	Physics	Source/Engine/ Physics/ SdfDamage.hpp	48
SdfPrimitiveInfo	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	137
SdfShape	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	82
SdfShapeComponent	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	127
SdfShapeType	Physics	Source/Engine/ Physics/ SdfPrimitives.hpp	16
SdfSmokeDesc	Physics	Source/Engine/ Physics/ SdfSmoke.hpp	16

SdfSmokeField	Physics	Source/Engine/ Physics/ SdfSmoke.hpp	33
SdfVolumeAsset	Physics	Source/Engine/ Physics/ SdfVolumeAsset.hpp	52
SdfVolumeHeader	Physics	Source/Engine/ Physics/ SdfVolumeAsset.hpp	26
SdfWorld	Physics	Source/Engine/ Physics/SdfWorld.hpp	60
SdfWorldBox	Physics	Source/Engine/ Physics/SdfWorld.hpp	45
SdfWorldCircle	Physics	Source/Engine/ Physics/SdfWorld.hpp	51
SdfWorldDesc	Physics	Source/Engine/ Physics/SdfWorld.hpp	31
SdfWorldImprint	Physics	Source/Engine/ Physics/SdfWorld.hpp	20
ShallowWaterGrid	Physics	Source/Engine/ Physics/Fluid.hpp	134
Storage	Physics	Source/Engine/ Physics/Cloth.hpp	155
Storage	Physics	Source/Engine/ Physics/ PhysicsJoints.hpp	642
SystemGraph	Physics	Source/Engine/ Physics/Physics.hpp	24
TilemapComponent	Physics	Source/Engine/ Physics/SdfWorld.hpp	15
Tracked	Physics	Source/Engine/ Physics/Fluid.hpp	255
TransportCapabilities	Physics	Source/Engine/ Physics/ TransportGroups.hpp	31
TransportGroupState	Physics	Source/Engine/ Physics/ TransportGroups.hpp	41
TransportGroupSystem	Physics	Source/Engine/ Physics/ TransportGroups.hpp	63
WaterView	Physics	Source/Engine/ Physics/Fluid.hpp	85

WheelSystem	Physics	Source/Engine/ Physics/ VehicleWheels.hpp	16

Engine/Platform (21)

Symbol	Kind	Header	Line
AchievementFlags	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	95
AchievementRequest	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	104
CloudSaveComponent	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	124
CloudSaveFlags	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	115
EgsStoreSystem	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	178
GogAchievementSystem	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	224
GogStorageSystem	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	258
PlatformAccountComponent	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	137
PlatformCommand	Platform	Source/Engine/ Platform/ PlatformService.hpp	35
PlatformCommandLog	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	46
PlatformCommandType	Platform	Source/Engine/ Platform/ PlatformService.hpp	24
PlatformLogEvent	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	35
PlatformLogType	Platform	Source/Engine/ Platform/	26

		PlatformSystems.hpp	
PlatformService	Platform	Source/Engine/ Platform/ PlatformService.hpp	76
PlatformServiceCallbacks	Platform	Source/Engine/ Platform/ PlatformService.hpp	51
PlatformServiceDesc	Platform	Source/Engine/ Platform/ PlatformService.hpp	71
PlatformServiceStats	Platform	Source/Engine/ Platform/ PlatformService.hpp	59
PlatformState	Platform	Source/Engine/ Platform/ PlatformService.hpp	132
StagedSave	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	204
StagedSave	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	291
SteamAchievementSystem	Platform	Source/Engine/ Platform/ PlatformSystems.hpp	148

Engine/RHI (193)

Symbol	Kind	Header	Line
AccessibilityCompositePush	RHI	Source/Engine/RHI/ Vulkan/ VulkanRHIInternal.hpp	282
Api	RHI	Source/Engine/RHI/ Vulkan/ VulkanLoader.hpp	123
Backend	RHI	Source/Engine/RHI/ RHI.hpp	19
BackendCapabilities	RHI	Source/Engine/RHI/ UpscalerCapabilities.hpp	28
BackendSelection	RHI	Source/Engine/RHI/ UpscalerBackend.hpp	28
CameraData	RHI	Source/Engine/RHI/ RenderItem.hpp	255

CameraDistortionFlags	RHI	Source/Engine/RHI/CameraStack.hpp	30
CameraDistortionSettings	RHI	Source/Engine/RHI/CameraStack.hpp	40
CameraStackDesc	RHI	Source/Engine/RHI/CameraStack.hpp	56
CameraStackFlags	RHI	Source/Engine/RHI/CameraStack.hpp	17
CameraStackSystem	RHI	Source/Engine/RHI/CameraStack.hpp	81
CameraStackSystem	RHI	Source/Engine/RHI/RenderGraph.hpp	17
CameraStackSystem	RHI	Source/Engine/RHI/RenderPasses.hpp	12
ChangeMaskRenderExtract	RHI	Source/Engine/RHI/ChangeMaskExtract.hpp	33
ClusterFrame	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	914
ClusterGridConstants	RHI	Source/Engine/RHI/GpuLighting.hpp	46
ClusterHeader	RHI	Source/Engine/RHI/GpuLighting.hpp	36
ColorGradingConstants	RHI	Source/Engine/RHI/RenderPasses.hpp	141
ColorGradingPass	RHI	Source/Engine/RHI/RenderPasses.hpp	149
ColorGradingSettings	RHI	Source/Engine/RHI/RenderPasses.hpp	129
CookedMaterialView	RHI	Source/Engine/RHI/MaterialAsset.hpp	152
CullFrame	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	944
CustomFrameGenerator	RHI	Source/Engine/RHI/CustomFrameGenerator.hpp	10
DeviceDesc	RHI	Source/Engine/RHI/RHI.hpp	49
DeviceInfo	RHI	Source/Engine/RHI/RHI.hpp	112

DirtyRange	RHI	Source/Engine/RHI/RenderItem.hpp	236
DlssAdapter	RHI	Source/Engine/RHI/DlssAdapter.hpp	33
DlssQualityMode	RHI	Source/Engine/RHI/DlssAdapter.hpp	9
DlssRuntimeAvailability	RHI	Source/Engine/RHI/DlssRuntime.hpp	9
DlssRuntimeProbe	RHI	Source/Engine/RHI/DlssRuntimeProbe.hpp	14
DrawBatch	RHI	Source/Engine/RHI/RenderItem.hpp	216
DrawIndexedIndirectCommand	RHI	Source/Engine/RHI/GpuVisibility.hpp	15
DynamicResolution	RHI	Source/Engine/RHI/DynamicResolution.hpp	19
DynamicResolutionSettings	RHI	Source/Engine/RHI/DynamicResolution.hpp	9
Entry	RHI	Source/Engine/RHI/MaterialPipelineCache.hpp	74
ExportStatus	RHI	Source/Engine/RHI/FrameExporter.hpp	100
Extent2D	RHI	Source/Engine/RHI/RHI.hpp	40
ExtractStats	RHI	Source/Engine/RHI/ChangeMaskExtract.hpp	21
FidelityFxRuntimeAvailability	RHI	Source/Engine/RHI/FidelityFxRuntime.hpp	25
FrameAccessibility	RHI	Source/Engine/RHI/RenderItem.hpp	277
FrameAccessibilityFlags	RHI	Source/Engine/RHI/RenderItem.hpp	264
FrameComposer	RHI	Source/Engine/RHI/FrameComposer.hpp	99
FrameComposerStats	RHI	Source/Engine/RHI/FrameComposer.hpp	80
FrameExportDesc	RHI	Source/Engine/RHI/FrameExporter.hpp	53

FrameExporter	RHI	Source/Engine/RHI/FrameExporter.hpp	112
FrameGenBackend	RHI	Source/Engine/RHI/UpscalerBackend.hpp	19
FrameGenerationInput	RHI	Source/Engine/RHI/FrameGeneration.hpp	10
FrameGenerationPolicy	RHI	Source/Engine/RHI/FrameGenerationPolicy.hpp	19
FrameGenerationStatus	RHI	Source/Engine/RHI/FrameGenerationPolicy.hpp	9
FramePacingConfig	RHI	Source/Engine/RHI/FramePacing.hpp	10
FramePacingState	RHI	Source/Engine/RHI/FramePacing.hpp	17
FramePacket	RHI	Source/Engine/RHI/RenderItem.hpp	387
FrameSink	RHI	Source/Engine/RHI/FrameExporter.hpp	85
FrameStats	RHI	Source/Engine/RHI/RHI.hpp	257
FrameTimestampIndex	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	76
Fsr2Adapter	RHI	Source/Engine/RHI/Fsr2Adapter.hpp	33
Fsr2Pass	RHI	Source/Engine/RHI/Fsr2Pass.hpp	10
Fsr2QualityMode	RHI	Source/Engine/RHI/Fsr2Adapter.hpp	9
Fsr3FrameGenerator	RHI	Source/Engine/RHI/Fsr3FrameGenerator.hpp	10
GiFrame	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	960
GpuCullingConstants	RHI	Source/Engine/RHI/GpuVisibility.hpp	28
GpuGiProbe	RHI	Source/Engine/RHI/GpuLighting.hpp	67
GpuGiVolumeHeader	RHI	Source/Engine/RHI/GpuLighting.hpp	57

GpuLight	RHI	Source/Engine/RHI/GpuLighting.hpp	18
GpuVendor	RHI	Source/Engine/RHI/UpscalerCapabilities.hpp	19
HashEntry	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	218
HiZPushConstants	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	66
IFrameGenerator	RHI	Source/Engine/RHI/FrameGeneration.hpp	27
ImageLayout	RHI	Source/Engine/RHI/RHI.hpp	218
ITemporalUpscaler	RHI	Source/Engine/RHI/TemporalUpscaler.hpp	28
JobSystem	RHI	Source/Engine/RHI/FrameComposer.hpp	24
KeySorter	RHI	Source/Engine/RHI/Sort.hpp	14
LoadRequest	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	241
LoadResult	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	248
LookupEntry	RHI	Source/Engine/RHI/ChangeMaskExtract.hpp	261
MaterialAssetError	RHI	Source/Engine/RHI/MaterialAsset.hpp	42
MaterialCompileOptions	RHI	Source/Engine/RHI/MaterialGraphCompiler.hpp	27
MaterialContract	RHI	Source/Engine/RHI/MaterialAsset.hpp	83
MaterialDomain	RHI	Source/Engine/RHI/MaterialGraphCompiler.hpp	21
MaterialGraph	RHI	Source/Engine/RHI/MaterialGraph.hpp	555
MaterialGraphCompi	RHI	Source/Engine/RHI/	58

ler		MaterialGraphCompiler.hpp	
MaterialGraphRequirements	RHI	Source/Engine/RHI/MaterialGraph.hpp	542
MaterialNode	RHI	Source/Engine/RHI/MaterialGraph.hpp	117
MaterialNodeFlags	RHI	Source/Engine/RHI/MaterialGraph.hpp	111
MaterialNodeKind	RHI	Source/Engine/RHI/MaterialGraph.hpp	29
MaterialParam	RHI	Source/Engine/RHI/MaterialGraph.hpp	99
MaterialParamKind	RHI	Source/Engine/RHI/MaterialGraph.hpp	88
MaterialPin	RHI	Source/Engine/RHI/MaterialGraph.hpp	129
MaterialPipeline	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	239
MaterialPipelineCache	RHI	Source/Engine/RHI/MaterialPipelineCache.hpp	38
MaterialPipelineCacheStats	RHI	Source/Engine/RHI/MaterialPipelineCache.hpp	18
MaterialShaderProgram	RHI	Source/Engine/RHI/MaterialGraphCompiler.hpp	39
MaterialValueType	RHI	Source/Engine/RHI/MaterialGraph.hpp	78
MeshArenaRange	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	207
MeshComputePush	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	132
MeshDesc	RHI	Source/Engine/RHI/Mesh.hpp	102
MeshDrawData	RHI	Source/Engine/RHI/Mesh.hpp	70
MeshDrawFlags	RHI	Source/Engine/RHI/Mesh.hpp	54
MeshDrawRecord	RHI	Source/Engine/RHI/	183

		Vulkan/ VulkanRHIInternal.hpp	
MeshFrameBuffers	RHI	Source/Engine/RHI/ Vulkan/ VulkanRHIInternal.hpp	150
MeshGraphicsPush	RHI	Source/Engine/RHI/ Vulkan/ VulkanRHIInternal.hpp	122
MeshHandle	RHI	Source/Engine/RHI/ Mesh.hpp	87
MeshIndirectGroup	RHI	Source/Engine/RHI/ Vulkan/ VulkanRHIInternal.hpp	198
MeshItemFlags	RHI	Source/Engine/RHI/ Mesh.hpp	62
MeshRenderItem	RHI	Source/Engine/RHI/ Mesh.hpp	114
MeshSkinRange	RHI	Source/Engine/RHI/ Mesh.hpp	46
MeshVertex	RHI	Source/Engine/RHI/ Mesh.hpp	17
MotionHistory2D	RHI	Source/Engine/RHI/ FrameComposer.hpp	33
MotionVectorItem	RHI	Source/Engine/RHI/ RenderItem.hpp	369
NativeDeviceHandles	RHI	Source/Engine/RHI/ RHI.hpp	709
NativeTextureView	RHI	Source/Engine/RHI/ RHI.hpp	224
OitPass	RHI	Source/Engine/RHI/ OitPass.hpp	11
PageSlot	RHI	Source/Engine/RHI/ VirtualTextureSystem.hpp	209
ParallelChunkRef	RHI	Source/Engine/RHI/ FrameComposer.hpp	301
PlatformKind	RHI	Source/Engine/RHI/ UpscalerCapabilities.hpp	11
PostFxAccessibilityEffect	RHI	Source/Engine/RHI/ PostProcess.hpp	33
PostFxConstants	RHI	Source/Engine/RHI/ RenderPasses.hpp	97

PostFxFlags	RHI	Source/Engine/RHI/PostProcess.hpp	11
PostFxPass	RHI	Source/Engine/RHI/RenderPasses.hpp	170
PostFxSettings	RHI	Source/Engine/RHI/PostProcess.hpp	56
PreUploadedPacket	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	1042
PushConstants	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	47
RenderBlendMode	RHI	Source/Engine/RHI/RHI.hpp	190
RenderCommandRecord	RHI	Source/Engine/RHI/RenderGraph.hpp	58
RenderFlags	RHI	Source/Engine/RHI/RenderItem.hpp	43
RenderGraph	RHI	Source/Engine/RHI/RenderGraph.hpp	135
RenderGraphContext	RHI	Source/Engine/RHI/RenderGraph.hpp	80
RenderGraphPass	RHI	Source/Engine/RHI/RenderGraph.hpp	122
RenderItem	RHI	Source/Engine/RHI/RenderItem.hpp	94
RenderPassDesc	RHI	Source/Engine/RHI/RenderGraph.hpp	110
RenderPassKind	RHI	Source/Engine/RHI/RenderGraph.hpp	27
RenderResourceKind	RHI	Source/Engine/RHI/RenderGraph.hpp	34
RenderTarget	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	250
RenderTargetDesc	RHI	Source/Engine/RHI/RHI.hpp	197
RenderTargetHandle	RHI	Source/Engine/RHI/RHI.hpp	207
RHI	RHI	Source/Engine/RHI/RHI.hpp	328
SdfCharacterRender	RHI	Source/Engine/RHI/	17

Cache		SdfCharacterRender.hpp	
SdfField	RHI	Source/Engine/RHI/FrameComposer.hpp	22
SdfGiVolume	RHI	Source/Engine/RHI/FrameComposer.hpp	23
SdfGpuBlock	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	98
SdfLightingFrame	RHI	Source/Engine/RHI/RenderItem.hpp	325
SdfLightingOccluder	RHI	Source/Engine/RHI/RenderItem.hpp	317
SdfShadowConstants	RHI	Source/Engine/RHI/RenderPasses.hpp	37
SdfShadowDesc	RHI	Source/Engine/RHI/RenderPasses.hpp	18
SdfShadowPass	RHI	Source/Engine/RHI/RenderPasses.hpp	66
SequenceFooter	RHI	Source/Engine/RHI/FrameExporter.hpp	42
SequenceHeader	RHI	Source/Engine/RHI/FrameExporter.hpp	31
ShaderPipelineDesc	RHI	Source/Engine/RHI/RHI.hpp	167
ShaderPipelineDomain	RHI	Source/Engine/RHI/RHI.hpp	161
ShaderPipelineHandle	RHI	Source/Engine/RHI/RHI.hpp	178
SkinnedVertex	RHI	Source/Engine/RHI/Mesh.hpp	30
Slot	RHI	Source/Engine/RHI/CameraStack.hpp	146
SpatialUpscaleMode	RHI	Source/Engine/RHI/RHI.hpp	90
StaticBakeCache	RHI	Source/Engine/RHI/StaticBake.hpp	58
StaticBakeRenderer	RHI	Source/Engine/RHI/StaticBakeRenderer.hpp	14
StaticBakeSettings	RHI	Source/Engine/RHI/StaticBake.hpp	10

StaticBakeStats	RHI	Source/Engine/RHI/StaticBake.hpp	47
StaticBakeTileKey	RHI	Source/Engine/RHI/StaticBake.hpp	17
StaticMeshRecord	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	213
TemporalResolvePass	RHI	Source/Engine/RHI/TemporalResolvePass.hpp	11
TemporalUpscaleInput	RHI	Source/Engine/RHI/TemporalUpscaler.hpp	11
TextureArray	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	222
TextureArrayDesc	RHI	Source/Engine/RHI/RHI.hpp	146
TextureArrayHandle	RHI	Source/Engine/RHI/RHI.hpp	304
Tile	RHI	Source/Engine/RHI/StaticBake.hpp	90
TintEntry	RHI	Source/Engine/RHI/ViewportPresenter.hpp	81
TransientRing	RHI	Source/Engine/RHI/RingBuffer.hpp	15
UiBackendInfo	RHI	Source/Engine/RHI/RHI.hpp	234
UpscalerBackend	RHI	Source/Engine/RHI/UpscalerBackend.hpp	9
UpscalerCaps	RHI	Source/Engine/RHI/RHI.hpp	98
ViewportMode	RHI	Source/Engine/RHI/ViewportModes.hpp	15
ViewportModeDescriptor	RHI	Source/Engine/RHI/ViewportModes.hpp	46
ViewportModeLegend	RHI	Source/Engine/RHI/ViewportModes.hpp	37
ViewportPresenter	RHI	Source/Engine/RHI/ViewportPresenter.hpp	29
ViewportPresenterSt	RHI	Source/Engine/RHI/	17

ats		ViewportPresenter.hpp	
VirtualFileSystem	RHI	Source/Engine/RHI/FrameExporter.hpp	15
VirtualFileSystem	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	22
VirtualTextureAddress	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	74
VirtualTextureCamera	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	65
VirtualTextureDesc	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	39
VirtualTextureHandle	RHI	Source/Engine/RHI/RenderItem.hpp	294
VirtualTexturePageLocation	RHI	Source/Engine/RHI/VirtualTextureGpu.hpp	11
VirtualTextureResource	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	424
VirtualTextureResourceDesc	RHI	Source/Engine/RHI/RHI.hpp	316
VirtualTextureRhiTarget	RHI	Source/Engine/RHI/VirtualTextureGpu.hpp	118
VirtualTextureSample	RHI	Source/Engine/RHI/VirtualTextureGpu.hpp	58
VirtualTextureStats	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	51
VirtualTextureSystem	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	117
VirtualTextureTableUpdate	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	107
VirtualTextureTileId	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	98
VirtualTextureTileUpload	RHI	Source/Engine/RHI/VirtualTextureSystem.hpp	83

VisibleSprite	RHI	Source/Engine/RHI/ViewportPresenter.hpp	87
VulkanRHI	RHI	Source/Engine/RHI/Vulkan/VulkanRHIInternal.hpp	440

Engine/Script (69)

Symbol	Kind	Header	Line
ActiveScript	Script	Source/Engine/Script/ScriptSystem.hpp	174
AotBuilder	Script	Source/Engine/Script/AotBuilder.hpp	71
AotBuildOptions	Script	Source/Engine/Script/AotBuilder.hpp	43
AotBuildResult	Script	Source/Engine/Script/AotBuilder.hpp	60
AotBuildState	Script	Source/Engine/Script/AotBuilder.hpp	19
AotGraphTranspileOptions	Script	Source/Engine/Script/AotGraphCompiler.hpp	13
AotHost	Script	Source/Engine/Script/AotHost.hpp	32
AotHost	Script	Source/Engine/Script/ScriptSystem.hpp	23
AotModule	Script	Source/Engine/Script/AotModule.hpp	22
AotProgramRef	Script	Source/Engine/Script/AotHost.hpp	22
AotTranspileOptions	Script	Source/Engine/Script/AotTranspiler.hpp	12
AudioSystem	Script	Source/Engine/Script/Vm.hpp	19
BoundProgram	Script	Source/Engine/Script/EcsBridge.hpp	101
CameraStackSystem	Script	Source/Engine/Script/	18

		ScriptSystem.hpp	
CameraStackSystem	Script	Source/Engine/ Script/Vm.hpp	18
Code	Script	Source/Engine/ Script/Vm.hpp	33
CompiledDebug	Script	Source/Engine/ Script/Bytecode.hpp	493
CompiledProgram	Script	Source/Engine/ Script/Bytecode.hpp	508
ComponentAccessor	Script	Source/Engine/ Script/EcsBridge.hpp	17
ComponentBinding	Script	Source/Engine/ Script/Bytecode.hpp	437
Context	Script	Source/Engine/ Script/AotAbi.hpp	112
DebugField	Script	Source/Engine/ Script/ ScriptDebug.hpp	123
DebugFrame	Script	Source/Engine/ Script/ ScriptDebug.hpp	103
DebugSectionHeader	Script	Source/Engine/ Script/Bytecode.hpp	181
DebugSpan	Script	Source/Engine/ Script/Bytecode.hpp	159
DebugSpanFlags	Script	Source/Engine/ Script/Bytecode.hpp	150
DebugVariable	Script	Source/Engine/ Script/Bytecode.hpp	172
EcsBridge	Script	Source/Engine/ Script/EcsBridge.hpp	26
FieldBinding	Script	Source/Engine/ Script/Bytecode.hpp	447
FunctionEntry	Script	Source/Engine/ Script/Bytecode.hpp	460
HostApi	Script	Source/Engine/ Script/AotAbi.hpp	54
Instruction	Script	Source/Engine/ Script/Bytecode.hpp	417
LogicCompareOp	Script	Source/Engine/ Script/ GraphCompiler.hpp	22

LogicCompileOptions	Script	Source/Engine/Script/GraphCompiler.hpp	775
LogicCompileResult	Script	Source/Engine/Script/GraphCompiler.hpp	782
LogicFieldBinding	Script	Source/Engine/Script/GraphCompiler.hpp	90
LogicGraph	Script	Source/Engine/Script/GraphCompiler.hpp	362
LogicLink	Script	Source/Engine/Script/GraphCompiler.hpp	351
LogicMathOp	Script	Source/Engine/Script/GraphCompiler.hpp	56
LogicNode	Script	Source/Engine/Script/GraphCompiler.hpp	333
LogicNodeKind	Script	Source/Engine/Script/GraphCompiler.hpp	111
Module	Script	Source/Engine/Script/AotAbi.hpp	379
NativeId	Script	Source/Engine/Script/Bytecode.hpp	18
OpCode	Script	Source/Engine/Script/Bytecode.hpp	235
PendingField	Script	Source/Engine/Script/Bytecode.hpp	574
PhysicsSystem	Script	Source/Engine/Script/ScriptSystem.hpp	17
PhysicsSystem	Script	Source/Engine/Script/Vm.hpp	17
PinType	Script	Source/Engine/Script/GraphCompiler.hpp	318
ProgramBuilder	Script	Source/Engine/Script/Bytecode.hpp	539
ProgramHeader	Script	Source/Engine/Script/Bytecode.hpp	470

ResolvedComponent	Script	Source/Engine/ Script/EcsBridge.hpp	93
ScriptComponent	Script	Source/Engine/ Script/ ScriptComponent.hpp	16
ScriptEvent	Script	Source/Engine/ Script/Bytecode.hpp	205
ScriptRuntime	Script	Source/Engine/ Script/ ScriptSystem.hpp	22
ScriptRuntime	Script	Source/Engine/ Script/ ScriptSystem.hpp	26
ScriptVm	Script	Source/Engine/ Script/Vm.hpp	121
ScriptVm	Script	Source/Engine/ Script/Vm.hpp	235
Slot	Script	Source/Engine/ Script/AotHost.hpp	94
Slot	Script	Source/Engine/ Script/ ScriptSystem.hpp	154
SourceLocation	Script	Source/Engine/ Script/ ScriptDebug.hpp	17
TextCompileOptions	Script	Source/Engine/ Script/TextScript.hpp	14
TransportGroupSystem	Script	Source/Engine/ Script/Vm.hpp	16
Vector2D	Script	Source/Engine/ Script/AotAbi.hpp	28
VmContext	Script	Source/Engine/ Script/Vm.hpp	98
VmDebugState	Script	Source/Engine/ Script/Vm.hpp	160
VmFault	Script	Source/Engine/ Script/Vm.hpp	31
VmHalt	Script	Source/Engine/ Script/Vm.hpp	133
VmPauseReason	Script	Source/Engine/ Script/Vm.hpp	124
VmStopInfo	Script	Source/Engine/ Script/Vm.hpp	140

Engine/Text (84)

Symbol	Kind	Header	Line
AssetSlot	Text	Source/Engine/Text/TextStore.hpp	218
BookAnchor	Text	Source/Engine/Text/BookAsset.hpp	107
BookAssetError	Text	Source/Engine/Text/BookAsset.hpp	140
BookAssetView	Text	Source/Engine/Text/BookAsset.hpp	127
BookBlockRecord	Text	Source/Engine/Text/BookAsset.hpp	60
BookBlockType	Text	Source/Engine/Text/BookAsset.hpp	22
BookBlockView	Text	Source/Engine/Text/BookAsset.hpp	117
BookCookError	Text	Source/Engine/Text/BookCook.hpp	13
BookHeader	Text	Source/Engine/Text/BookAsset.hpp	35
BookInline	Text	Source/Engine/Text/BookAsset.hpp	97
BookPage	Text	Source/Engine/Text/BookSession.hpp	25
BookPageSegment	Text	Source/Engine/Text/BookSession.hpp	35
BookReader	Text	Source/Engine/Text/BookAsset.hpp	159
BookSession	Text	Source/Engine/Text/BookSession.hpp	46
BookSpan	Text	Source/Engine/Text/BookAsset.hpp	77
BookStyle	Text	Source/Engine/Text/BookAsset.hpp	86
BookViewport	Text	Source/Engine/Text/BookSession.hpp	19
ChainedShaper	Text	Source/Engine/Text/TextShaper.hpp	133
CodepointShaper	Text	Source/Engine/Text/TextShaper.hpp	99

Entry	Text	Source/Engine/Text/TextShaper.hpp	247
FontCandidate	Text	Source/Engine/Text/TextFontStack.hpp	24
FontSlot	Text	Source/Engine/Text/TextFontStack.hpp	59
FormatEntry	Text	Source/Engine/Text/TextService.hpp	583
FormattedLayoutEntry	Text	Source/Engine/Text/TextService.hpp	566
FormattedLayoutKey	Text	Source/Engine/Text/TextService.hpp	556
HarfBuzzShaper	Text	Source/Engine/Text/TextShaper.hpp	110
Impl	Text	Source/Engine/Text/TextShaper.hpp	128
ITextShaper	Text	Source/Engine/Text/TextShaper.hpp	89
LayoutEntry	Text	Source/Engine/Text/TextService.hpp	548
LayoutKey	Text	Source/Engine/Text/TextService.hpp	528
MappedTextAsset	Text	Source/Engine/Text/TextAssetFile.hpp	17
MarkupStackEntry	Text	Source/Engine/Text/TextMarkup.hpp	71
MarkupVariantEntry	Text	Source/Engine/Text/TextService.hpp	574
ScriptBackend	Text	Source/Engine/Text/TextShapeSystem.hpp	124
ScriptShaper	Text	Source/Engine/Text/TextShaper.hpp	207
ShapedGlyph	Text	Source/Engine/Text/TextShaper.hpp	56
ShapedGlyphHit	Text	Source/Engine/Text/TextShapedLayout.hpp	153
ShapedLayoutResult	Text	Source/Engine/Text/TextShapedLayout.hpp	26
ShapedLine	Text	Source/Engine/Text/	18

		TextShapedLayout.hpp	
StyledAsset	Text	Source/Engine/Text/TextService.hpp	595
TextAction	Text	Source/Engine/Text/TextStyle.hpp	61
TextActionKind	Text	Source/Engine/Text/TextStyle.hpp	51
TextAnimationParams	Text	Source/Engine/Text/TextRender.hpp	29
TextAssetBlock	Text	Source/Engine/Text/TextAssetFormat.hpp	34
TextAssetHeader	Text	Source/Engine/Text/TextAssetFormat.hpp	20
TextAssetRef	Text	Source/Engine/Text/TextAsset.hpp	14
TextAssetStatus	Text	Source/Engine/Text/TextAssetFormat.hpp	63
TextAssetView	Text	Source/Engine/Text/TextAssetFormat.hpp	42
TextBlock	Text	Source/Engine/Text/TextAsset.hpp	37
TextDirection	Text	Source/Engine/Text/TextShaper.hpp	21
TextFontStack	Text	Source/Engine/Text/TextFontStack.hpp	54
TextFormatArg	Text	Source/Engine/Text/TextFormat.hpp	45
TextFormatStatus	Text	Source/Engine/Text/TextFormat.hpp	33
TextGradientParams	Text	Source/Engine/Text/TextRender.hpp	22
TextInlineHit	Text	Source/Engine/Text/TextLayout.hpp	204
TextInlineKind	Text	Source/Engine/Text/TextStyle.hpp	73
TextInlineObject	Text	Source/Engine/Text/TextStyle.hpp	83
TextInlineSprite	Text	Source/Engine/Text/TextStyle.hpp	99
TextLayoutParams	Text	Source/Engine/Text/	54

		TextLayout.hpp	
TextLayoutResult	Text	Source/Engine/Text/TextLayout.hpp	72
TextLine	Text	Source/Engine/Text/TextLayout.hpp	22
TextMarkupStatus	Text	Source/Engine/Text/TextMarkup.hpp	16
TextRenderEffects	Text	Source/Engine/Text/TextRender.hpp	44
TextRenderParams	Text	Source/Engine/Text/TextRender.hpp	57
TextRenderStats	Text	Source/Engine/Text/TextRender.hpp	76
TextRenderSystemStats	Text	Source/Engine/Text/TextRenderSystem.hpp	19
TextRun	Text	Source/Engine/Text/TextShaper.hpp	47
TextRuntime	Text	Source/Engine/Text/TextRuntime.hpp	17
TextScript	Text	Source/Engine/Text/TextShaper.hpp	30
TextService	Text	Source/Engine/Text/TextService.hpp	18
TextShapeCache	Text	Source/Engine/Text/TextShapeCache.hpp	59
TextShapeHandle	Text	Source/Engine/Text/TextShapeCache.hpp	19
TextShapeKey	Text	Source/Engine/Text/TextShapeCache.hpp	32
TextShapeParams	Text	Source/Engine/Text/TextShaper.hpp	71
TextShapeResult	Text	Source/Engine/Text/TextShaper.hpp	82
TextShapeSystemStats	Text	Source/Engine/Text/TextShapeSystem.hpp	18
TextShapeView	Text	Source/Engine/Text/TextShapeCache.hpp	50
TextSpan	Text	Source/Engine/Text/TextStyle.hpp	40
TextStore	Text	Source/Engine/Text/	19

		TextStore.hpp	
TextStyle	Text	Source/Engine/Text/TextStyle.hpp	27
TextStyledBlock	Text	Source/Engine/Text/TextMarkup.hpp	30
TextStyleFlags	Text	Source/Engine/Text/TextStyle.hpp	12
UnicodeBoundary	Text	Source/Engine/Text/UnicodeBreak.hpp	13
VerticalFormEntry	Text	Source/Engine/Text/TextVertical.hpp	16

Engine/UI (115)

Symbol	Kind	Header	Line
AccessibilityEditorSettings	UI	Source/Engine/UI/Accessibility.hpp	251
AccessibilityEffectKind	UI	Source/Engine/UI/AccessibilityEffects.hpp	26
AccessibilityEffectProfile	UI	Source/Engine/UI/AccessibilityEffects.hpp	84
AccessibilityEffectRegistry	UI	Source/Engine/UI/AccessibilityEffects.hpp	122
AccessibilityEffectTag	UI	Source/Engine/UI/AccessibilityEffects.hpp	58
AccessibilityExtensionRegistry	UI	Source/Engine/UI/AccessibilityEffects.hpp	103
AccessibilityFlags	UI	Source/Engine/UI/Accessibility.hpp	209
AccessibilityKernel	UI	Source/Engine/UI/Accessibility.hpp	305
AccessibilityOverlay	UI	Source/Engine/UI/AccessibilityOverlay.hpp	30
AccessibilityPaletteExtension	UI	Source/Engine/UI/AccessibilityEffects.hpp	42
AccessibilityPresent	UI	Source/Engine/UI/	234

ationSettings		Accessibility.hpp	
AccessibilityProfile	UI	Source/Engine/UI/Accessibility.hpp	273
AccessibilitySettings	UI	Source/Engine/UI/Accessibility.hpp	218
AccessibilitySlot	UI	Source/Engine/UI/AccessibilityTheme.hpp	61
AccessibilityTemperatureBand	UI	Source/Engine/UI/AccessibilityEffects.hpp	49
AccessibilityTextStyle	UI	Source/Engine/UI/AccessibilityTheme.hpp	82
AccessibilityTheme	UI	Source/Engine/UI/AccessibilityTheme.hpp	92
AccessibilityThemeRegistry	UI	Source/Engine/UI/AccessibilityTheme.hpp	116
AlcoholEffect	UI	Source/Engine/UI/AccessibilityEffects.hpp	63
AlcoholEffectLayer	UI	Source/Engine/UI/AccessibilityEffects.hpp	77
AudioSystem	UI	Source/Engine/UI/Accessibility.hpp	27
AvSpeechTts	UI	Source/Engine/UI/Tts.hpp	60
BuildStatus	UI	Source/Engine/UI/PinkUI.hpp	1446
CardBoardComponent	UI	Source/Engine/UI/CardBoard.hpp	49
CardBoardSystem	UI	Source/Engine/UI/CardBoard.hpp	74
ColorMatrix3	UI	Source/Engine/UI/Accessibility.hpp	43
ColorVisionDeficiency	UI	Source/Engine/UI/Accessibility.hpp	35
Dropdown	UI	Source/Engine/UI/PinkUI.hpp	535
FlashGuard	UI	Source/Engine/UI/	60

		Accessibility.hpp	
FocusDirection	UI	Source/Engine/UI/PinkUI.hpp	212
FocusPolicy	UI	Source/Engine/UI/PinkUI.hpp	188
FontAtlas	UI	Source/Engine/UI/FontAtlas.hpp	101
FontAtlasHeader	UI	Source/Engine/UI/FontAtlas.hpp	45
FontAtlasSet	UI	Source/Engine/UI/FontAtlas.hpp	167
FontBuildOptions	UI	Source/Engine/UI/FontAtlas.hpp	25
FontBuildOptions	UI	Source/Engine/UI/FontBuilder.hpp	15
FontCoverage	UI	Source/Engine/UI/FontCoverage.hpp	21
FontFamilyFile	UI	Source/Engine/UI/FontAtlas.hpp	74
FontFamilyManifest	UI	Source/Engine/UI/FontFamilyAsset.hpp	17
FontPixelFormat	UI	Source/Engine/UI/FontAtlas.hpp	37
FontStyle	UI	Source/Engine/UI/FontAtlas.hpp	65
FrameOutput	UI	Source/Engine/UI/PinkUI.hpp	1421
GlyphRecord	UI	Source/Engine/UI/FontAtlas.hpp	80
GridView	UI	Source/Engine/UI/PinkUI.hpp	425
HyperlinkRange	UI	Source/Engine/UI/Hypertext.hpp	51
HypertextComponent	UI	Source/Engine/UI/Hypertext.hpp	32
HypertextFlags	UI	Source/Engine/UI/Hypertext.hpp	21
HypertextPool	UI	Source/Engine/UI/Hypertext.hpp	74
HypertextView	UI	Source/Engine/UI/Hypertext.hpp	67

ITextShaper	UI	Source/Engine/UI/PinkUI.hpp	34
LayoutCache	UI	Source/Engine/UI/PinkUI.hpp	741
LazyFontFamily	UI	Source/Engine/UI/FontFamilyAsset.hpp	101
ListView	UI	Source/Engine/UI/PinkUI.hpp	561
LocalizationTable	UI	Source/Engine/UI/Accessibility.hpp	25
LocalizationTable	UI	Source/Engine/UI/Hypertext.hpp	16
LocalizationTable	UI	Source/Engine/UI/PinkUI.hpp	40
NarrativeArchetype	UI	Source/Engine/UI/NarrativeTree.hpp	23
NarrativeEvent	UI	Source/Engine/UI/Accessibility.hpp	84
NarrativeGroup	UI	Source/Engine/UI/NarrativeTree.hpp	37
NarrativeLocale	UI	Source/Engine/UI/NarrativeTree.hpp	17
NarrativeLog	UI	Source/Engine/UI/Accessibility.hpp	100
NarrativeNode	UI	Source/Engine/UI/NarrativeTree.hpp	47
NarrativeTree	UI	Source/Engine/UI/NarrativeTree.hpp	55
NvgPattern	UI	Source/Engine/UI/AccessibilityEffects.hpp	33
OledShiftState	UI	Source/Engine/UI/PinkUI.hpp	464
PinkSdfDraw	UI	Source/Engine/UI/PinkSdf.hpp	74
PinkSdfFlags	UI	Source/Engine/UI/PinkSdf.hpp	24
PinkSdfGpuPrimitive	UI	Source/Engine/UI/PinkSdf.hpp	51
PinkSdfPrimitive	UI	Source/Engine/UI/PinkSdf.hpp	33
PinkSdfShape	UI	Source/Engine/UI/	15

		PinkSdf.hpp	
RadialProgress	UI	Source/Engine/UI/ PinkUI.hpp	585
RetainedFont	UI	Source/Engine/UI/ FontAtlas.hpp	293
SapiTts	UI	Source/Engine/UI/ Tts.hpp	30
ScrollView	UI	Source/Engine/UI/ PinkUI.hpp	492
SlotComponent	UI	Source/Engine/UI/ CardBoard.hpp	27
SlotFlags	UI	Source/Engine/UI/ CardBoard.hpp	15
SlotLayout	UI	Source/Engine/UI/ CardBoard.hpp	65
SoundCueCategory	UI	Source/Engine/UI/ SoundCues.hpp	26
SoundCueDef	UI	Source/Engine/UI/ SoundCues.hpp	62
SoundCueRegistry	UI	Source/Engine/UI/ SoundCues.hpp	71
SoundTrigger	UI	Source/Engine/UI/ Accessibility.hpp	129
StateTrack	UI	Source/Engine/UI/ Accessibility.hpp	424
SubtitleLine	UI	Source/Engine/UI/ Accessibility.hpp	140
SubtitleQueue	UI	Source/Engine/UI/ Accessibility.hpp	164
SystemContext	UI	Source/Engine/UI/ PinkUI.hpp	29
TabBar	UI	Source/Engine/UI/ PinkUI.hpp	516
TextAlign	UI	Source/Engine/UI/ PinkUI.hpp	223
TextShapeParams	UI	Source/Engine/UI/ PinkUI.hpp	35
TextVerticalAlign	UI	Source/Engine/UI/ PinkUI.hpp	230
TimedSoundCue	UI	Source/Engine/UI/ Accessibility.hpp	153

TooltipPlacer	UI	Source/Engine/UI/PinkUI.hpp	596
TtsBackend	UI	Source/Engine/UI/Accessibility.hpp	28
TtsBackend	UI	Source/Engine/UI/Tts.hpp	19
TtsPriority	UI	Source/Engine/UI/Tts.hpp	11
UiPalette	UI	Source/Engine/UI/UiTheme.hpp	41
UiPaletteSlot	UI	Source/Engine/UI/UiTheme.hpp	18
UiThemeAsset	UI	Source/Engine/UI/UiTheme.hpp	46
VirtualFileSystem	UI	Source/Engine/UI/FontAtlas.hpp	19
VoiceCaptionEvent	UI	Source/Engine/UI/Accessibility.hpp	194
VoiceCaptionProvider	UI	Source/Engine/UI/Accessibility.hpp	202
WidgetCanvas	UI	Source/Engine/UI/Accessibility.hpp	31
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	291
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	484
WidgetCanvas	UI	Source/Engine/UI/PinkUI.hpp	608
WidgetCanvasSystem	UI	Source/Engine/UI/PinkUI.hpp	1459
WidgetComponent	UI	Source/Engine/UI/PinkUI.hpp	252
WidgetComponentFlags	UI	Source/Engine/UI/PinkUI.hpp	264
WidgetDesc	UI	Source/Engine/UI/PinkUI.hpp	310
WidgetFlags	UI	Source/Engine/UI/PinkUI.hpp	146
WidgetHandle	UI	Source/Engine/UI/PinkUI.hpp	81
WidgetHyperlink	UI	Source/Engine/UI/	301

		PinkUI.hpp	
WidgetKind	UI	Source/Engine/UI/ PinkUI.hpp	100
WidgetLayoutAsset	UI	Source/Engine/UI/ WidgetLayoutAsset.hp p	15
WidgetPaintContext	UI	Source/Engine/UI/ PinkUI.hpp	292
WidgetPaintContext	UI	Source/Engine/UI/ PinkUI.hpp	392

Engine/Video (28)

Symbol	Kind	Header	Line
AssetDatabase	Video	Source/Engine/ Video/ VideoStream.hpp	28
AudioPumpStatus	Video	Source/Engine/ Video/ VideoAudioStream.hp p	15
ClusterIndexEntry	Video	Source/Engine/ Video/ WebmDemuxer.hpp	210
ClusterScan	Video	Source/Engine/ Video/ WebmDemuxer.hpp	150
ContainerKind	Video	Source/Engine/ Video/ WebmDemuxer.hpp	225
DecodeStatus	Video	Source/Engine/ Video/VideoTypes.hpp	33
DemuxStatus	Video	Source/Engine/ Video/ WebmDemuxer.hpp	74
Element	Video	Source/Engine/ Video/ WebmDemuxer.hpp	127
FrameBuffer	Video	Source/Engine/ Video/ VideoPlayer.hpp	160
FrameSlot	Video	Source/Engine/ Video/ VideoPlayer.hpp	167

LacedAudioPacket	Video	Source/Engine/ Video/ WebmDemuxer.hpp	201
MappedFile	Video	Source/Engine/ Video/ VideoPlayer.hpp	24
RHI	Video	Source/Engine/ Video/ VideoPlayer.hpp	29
SystemGraph	Video	Source/Engine/ Video/ VideoStream.hpp	24
VideoAudioDecoder	Video	Source/Engine/ Video/ VideoAudioDecoder.hpp	15
VideoAudioStream	Video	Source/Engine/ Video/ VideoAudioStream.hpp	27
VideoPlayer	Video	Source/Engine/ Video/ VideoPlayer.hpp	35
VideoStreamComponent	Video	Source/Engine/ Video/ VideoStream.hpp	36
VideoStreamFlags	Video	Source/Engine/ Video/VideoTypes.hpp	20
VideoSystem	Video	Source/Engine/ Video/ VideoStream.hpp	56
VirtualFileSystem	Video	Source/Engine/ Video/ VideoStream.hpp	20
Vp9Backend	Video	Source/Engine/ Video/ Vp9Backend.hpp	30
WebmAudioPacket	Video	Source/Engine/ Video/ WebmDemuxer.hpp	68
WebmAudioTrackInfo	Video	Source/Engine/ Video/ WebmDemuxer.hpp	43
WebmDemuxer	Video	Source/Engine/ Video/	82

		WebmDemuxer.hpp	
WebmFrame	Video	Source/Engine/ Video/ WebmDemuxer.hpp	54
WebmTrackInfo	Video	Source/Engine/ Video/ WebmDemuxer.hpp	15
YuvFrameView	Video	Source/Engine/ Video/ Vp9Backend.hpp	12

Engine/Weather (12)

Symbol	Kind	Header	Line
ClothSystem	Weather	Source/Engine/ Weather/Weather.hpp	13
WeatherAtmosphere Pass	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	128
WeatherContextCom ponent	Weather	Source/Engine/ Weather/Weather.hpp	32
WeatherDesc	Weather	Source/Engine/ Weather/Weather.hpp	91
WeatherFlags	Weather	Source/Engine/ Weather/Weather.hpp	19
WeatherGpuConstan ts	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	46
WeatherGpuResourc es	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	18
WeatherGpuStats	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	68
WeatherPreconciliat edDesc	Weather	Source/Engine/ Weather/ WeatherPreconciliated .hpp	32
WeatherPreconciliat edSystem	Weather	Source/Engine/ Weather/ WeatherPreconciliated	78

		.hpp	
WeatherSystem	Weather	Source/Engine/ Weather/Weather.hpp	99
WeatherWindField	Weather	Source/Engine/ Weather/Weather.hpp	75

21. Appendix B - macros and platform defines

Macro	Header	Line
FBY_ALCOHOL_EFFECTS	Source/Engine/Core/ PluginAPI.h	105
FBY_AOT_EXPORT	Source/Engine/Script/ AotAbi.hpp	396
FBY_AOT_EXPORT	Source/Engine/Script/ AotAbi.hpp	398
FBY_ASSERT	Source/Engine/Core/ Assert.hpp	13
FBY_ASSERT	Source/Engine/Core/ Assert.hpp	31
FBY_ASSERT_MSG	Source/Engine/Core/ Assert.hpp	22
FBY_ASSERT_MSG	Source/Engine/Core/ Assert.hpp	32
FBY_DEFINE_PLUGIN	Source/Engine/Core/ Plugin.hpp	55
FBY_DETAIL_CONCAT	Source/Engine/Core/ Definitions.hpp	263
FBY_DETAIL_CONCAT_INNER	Source/Engine/Core/ Definitions.hpp	262
FBY_EFFECT_TAGS	Source/Engine/Core/ PluginAPI.h	104
FBY_FIELD	Source/Engine/Core/ Mirror.hpp	288
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/ Fluid.hpp	16
FBY_FLUID_HAS_MDSPAN	Source/Engine/Physics/ Fluid.hpp	18
FBY_HAS_MDSPAN	Source/Engine/ECS/ Skeleton.hpp	18
FBY_HAS_MDSPAN	Source/Engine/ECS/	20

	Skeleton.hpp	
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/Mirror.hpp	105
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/Mirror.hpp	108
FBY_HAS_STATIC_REFLECTION	Source/Engine/Core/Mirror.hpp	112
FBY_HAS_STD_FLAT_MAP	Source/Engine/Core/FlatMap.hpp	19
FBY_PALETTE_EXTENSION_COLORS	Source/Engine/Core/PluginAPI.h	103
FBY_PLATFORM_ABI_VERSION	Source/Plugins/PlatformService/PlatformAPI.h	11
FBY_PLATFORM_CALL	Source/Plugins/PlatformService/PlatformAPI.h	15
FBY_PLATFORM_CALL	Source/Plugins/PlatformService/PlatformAPI.h	18
FBY_PLATFORM_EXPORT	Source/Plugins/PlatformService/PlatformAPI.h	14
FBY_PLATFORM_EXPORT	Source/Plugins/PlatformService/PlatformAPI.h	17
FBY_PLUGIN_ABI_VERSION	Source/Engine/Core/PluginAPI.h	30
FBY_PLUGIN_API_H	Source/Engine/Core/PluginAPI.h	18
FBY_PLUGIN_COMPONENT	Source/Engine/Core/Plugin.hpp	52
FBY_PLUGIN_ENTRY	Source/Engine/Core/PluginAPI.h	214
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginAPI.h	33
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginAPI.h	35
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginAPI.h	37
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginApi.hpp	11
FBY_PLUGIN_EXPORT	Source/Engine/Core/PluginApi.hpp	13
FBY_PLUGIN_INFO_SYMBOL	Source/Engine/Core/PluginAPI.h	206

FBY_PLUGIN_LOAD_SYMBOL	Source/Engine/Core/PluginAPI.h	207
FBY_PLUGIN_UNLOAD_SYMBOL	Source/Engine/Core/PluginAPI.h	208
FBY_PROJECT_ASSETS	Source/Engine/Assets/ProjectScaffold.hpp	129
FBY_PROJECT_MAX_RENDER_ITEMS	Source/Engine/Assets/ProjectScaffold.hpp	122
FBY_REFLECT_COMPONENT	Source/Engine/Core/Mirror.hpp	295
FBY_REFLECT_COMPONENT_EMPTY	Source/Engine/Core/Mirror.hpp	311
FBY_REGISTER_TYPE	Source/Engine/Core/Definitions.hpp	266
FBY_RELEVANCE_HAS_MDSPAN	Source/Engine/Network/RelevanceGrid.hpp	19
FBY_RELEVANCE_HAS_MDSPAN	Source/Engine/Network/RelevanceGrid.hpp	21
FBY_RESTRICT	Source/Engine/Core/Definitions.hpp	26
FBY_RESTRICT	Source/Engine/Core/Definitions.hpp	28
FBY_SDF_SHAPE_FIELD	Source/Engine/Physics/SdfPrimitives.hpp	172
FBY_TEMPERATURE_BANDS	Source/Engine/Core/PluginAPI.h	106
FBY_THEME_COLOR_COUNT	Source/Engine/Core/PluginAPI.h	84
FBY_TYPE_NAME	Source/Engine/Core/Definitions.hpp	282
FBY_UI_HAS_MDSPAN	Source/Engine/UI/PinkUI.hpp	22
FBY_UI_HAS_MDSPAN	Source/Engine/UI/PinkUI.hpp	24
FBY_UNREACHABLE	Source/Engine/Core/Assert.hpp	45
FBY_VERIFY	Source/Engine/Core/Assert.hpp	36
FBY_VK_DECLARE	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	127
FBY_VK_DEVICE_FUNCTIONS	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	36
FBY_VK_GLOBAL_FUNCTIO	Source/Engine/RHI/Vulkan/	15

NS	VulkanLoader.hpp	
FBY_VK_INSTANCE_FUNCTIONS	Source/Engine/RHI/Vulkan/VulkanLoader.hpp	20
FBY_VOICE_API_H	Source/Engine/Core/VoiceAPI.h	12
FBY_VOICE_API_VERSION	Source/Engine/Core/VoiceAPI.h	21
FBY_VOICE_FLAG_ALLOW_DOWNLINK	Source/Engine/Core/VoiceAPI.h	53
FBY_VOICE_FLAG_ALLOW_UPLINK	Source/Engine/Core/VoiceAPI.h	52
FBY_VOICE_FLAG_COSMETIC	Source/Engine/Core/VoiceAPI.h	50
FBY_VOICE_FLAG_MODERATION	Source/Engine/Core/VoiceAPI.h	51
FBY_VOICE_SLOT_COUNT	Source/Engine/Core/VoiceAPI.h	22
FORCEINLINE	Source/Engine/Core/Definitions.hpp	17
FORCEINLINE	Source/Engine/Core/Definitions.hpp	19
FORCEINLINE	Source/Engine/Core/Definitions.hpp	21